

Bridging Stochastic Control and Deep Hedging: Structural Priors for No-Transaction Band Networks

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March 2026

Abstract

This paper studies the problem of hedging and pricing a European call option under proportional transaction costs, from two complementary perspectives. We first derive the optimal hedging strategy under CARA utility, following the stochastic control framework of [Davis et al. \(1993\)](#), characterising the no-transaction band via the Hamilton–Jacobi–Bellman Quasi-Variational Inequality (HJBQVI) and the Whalley–Wilmott asymptotic approximation. We then adopt a deep hedging approach, proposing two architectures that build on the No-Transaction Band Network of [Imaki et al. \(2023\)](#): NTBN- Δ , which makes delta-centring explicit, and WW-NTBN, which incorporates the Whalley–Wilmott formula as a structural prior on the bandwidth and replaces the hard clamp with a differentiable soft clamp. Numerical experiments show that WW-NTBN converges faster, matches the stochastic control no-transaction bands more closely, and generalises well across transaction cost regimes. We further apply both frameworks to the bull call spread, documenting the breakdown of price linearity under transaction costs.

Key Words : Stochastic control, Deep Learning, Option Hedging, Transaction costs

1 Introduction

1.1 Motivation

The Black-Scholes model ([Black and Scholes, 1973](#)) provides the standard framework for option pricing and hedging. Its central result, that a continuously rebalanced portfolio in the underlying asset perfectly replicates any European payoff, underpins much of modern derivatives theory. Yet continuous rebalancing is a mathematical idealisation. Every transaction in the underlying asset incurs a cost: bid-ask spreads, exchange fees,

market impact, etc. Even modest proportional costs, applied to the rapid rebalancing that perfect replication demands, accumulate into charges that erode the value of any hedging strategy. The assumptions of Black-Scholes are therefore not just imprecise but qualitatively misleading in a market where transaction costs are present.

The consequences go further than a simple reduction in profitability. Transaction costs fundamentally change the structure of optimal strategies. Under frictions, continuous delta-hedging is no longer optimal; it is dominated by strategies that trade only when the portfolio drifts sufficiently far from the target position. This inactivity is not a failure of the hedger but a rational response: the cost of rebalancing must be weighed against the utility loss from holding a suboptimal position. The resulting no-transaction band, a region of the state space where it is optimal to do nothing, is the defining structural feature of optimal hedging under proportional transaction costs and one of the central objects of study in this paper.

From a theoretical perspective, the no-transaction band characterises the geometry of the optimal control problem and connects the hedging strategy to indifference pricing in a precise way. From a practical perspective, it determines how often and how aggressively a hedger should trade, directly controlling the running cost of maintaining a derivatives book.

1.2 Related Work

The problem of option replication under proportional transaction costs was first studied systematically by [Leland \(1985\)](#), who proposed adjusting the Black-Scholes volatility upward to absorb the expected cost of discrete rebalancing. While widely used in practice, this approach is heuristic: it does not arise from a utility-maximisation argument and does not characterise the optimal trading policy.

A rigorous treatment using stochastic control was initiated by [Hodges and Neuberger \(1989\)](#), who introduced the utility indifference pricing approach and established the existence of a no-transaction region. This framework was subsequently formalised by [Davis et al. \(1993\)](#), who formulated the problem as a singular stochastic control problem, derived the associated Hamilton-Jacobi-Bellman quasi-variational inequality (HJBQVI), and provided the first analytical characterisation of the no-transaction band boundaries. Related work by [Boyle and Vorst \(1992\)](#) and [Edirisinghe et al. \(1993\)](#) addressed discrete-time replication with transaction costs from a different angle, deriving bounds on replication costs without a utility framework. The connection between transaction costs and indifference pricing is surveyed in [Henderson and Hobson \(2004\)](#).

A key simplification was achieved by [Whalley and Wilmott \(1997\)](#), who derived an explicit asymptotic formula for the width of the no-transaction band in the limit of small transaction costs. Their result expresses the half-bandwidth as a function of the Black-

Scholes gamma, the risk aversion and the transaction cost level, giving an analytically tractable approximation that remains accurate for small to moderate costs. Related asymptotic results for small transaction costs were obtained by [Kallsen and Muhle-Karbe \(2015\)](#) in a more general semimartingale setting. This formula plays a central role in our work.

The deep hedging framework of [Buehler et al. \(2019\)](#) recast option hedging as a learning problem. A neural network is trained end-to-end to minimise a convex risk measure over simulated market trajectories, without requiring an explicit model for the price dynamics. This approach handles transaction costs, path-dependent payoffs, and portfolio-level constraints within a single unified framework, and has been shown to produce competitive hedging strategies across a range of market conditions. Extensions to rough volatility dynamics were studied by [Horvath et al. \(2021\)](#).

Within this framework, [Imaki et al. \(2023\)](#) proposed the No-Transaction Band Network (NTBN), which outputs the lower and upper boundaries of a no-transaction band at each time step rather than a direct hedge position. The hedge is updated only when the previous position falls outside this band, explicitly encoding the structural feature known from stochastic control theory. This design improves training stability and performance in high-friction regimes. In their experimental implementation, [Imaki et al.](#) centre the band around the Black-Scholes delta by adding and subtracting the network outputs from Δ_{BS} , though they do not present this as an explicit architectural choice or provide theoretical motivation for it. Beyond delta-centring, significant structural information available from the Whalley–Wilmott approximation, in particular the correct scaling of the bandwidth with transaction cost and risk aversion, is not incorporated into the architecture.

1.3 Contributions

This paper makes the following contributions.

Unified theoretical exposition. We provide a self-contained derivation of the optimal hedging problem under proportional transaction costs, following the stochastic control framework of [Davis et al. \(1993\)](#) specialised to CARA utility. We derive the HJBQVI, characterise the no-transaction band, and present the Whalley–Wilmott asymptotic expansion with a sketch of the key derivation steps in the appendix. While the individual results are established in the literature, we assemble them into a unified treatment that makes the connection to the deep hedging architectures below explicit. We also treat the bull call spread as a case study illustrating the breakdown of price linearity under transaction costs, a consequence of the no-transaction band structure that is often overlooked in practice.

NTBN- Δ : making delta-centring explicit. The original NTBN of Imaki et al. (2023) centres its band around Δ_{BS} in practice, but treats this as an unmarked implementation detail rather than an architectural principle. We formalise this choice as an explicit design decision, motivated directly by the stochastic control analysis: the frictionless optimal holding (21) shows that the natural centre of the no-transaction band is the Black-Scholes delta. The resulting architecture, NTBN- Δ , makes the delta-centring transparent, provides a clear theoretical justification for it, and serves as the principled baseline on which WW-NTBN is built.

WW-NTBN: a Whalley–Wilmott guided architecture. Building on NTBN- Δ , we propose the Whalley–Wilmott No-Transaction Band Network (WW-NTBN), which incorporates two further improvements. First, the initial half-bandwidth is set to the Whalley–Wilmott analytical approximation

$$h_{\text{WW}} = \left(\frac{3\lambda S \Gamma_{\text{BS}}^2}{2\gamma} \right)^{1/3}, \quad (1)$$

and the network learns only a residual correction on top of this prior. This gives the network the correct scaling with transaction cost λ and risk aversion γ at initialisation, accelerating convergence and improving generalisation across parameter regimes. Second, the hard clamp used to enforce the band is replaced by a differentiable soft clamp

$$\text{softclamp}(x, \ell, u) = \frac{\ell + u}{2} + \frac{u - \ell}{2} \tanh\left(\beta \frac{x - \frac{\ell+u}{2}}{\frac{u-\ell}{2}}\right), \quad (2)$$

which provides non-zero gradients even when the previous hedge lies outside the band, removing a source of vanishing gradients present in the standard NTBN. We note that Imaki et al. (2023) themselves flag the vanishing gradient issue in the clamped region as a training tip, but do not propose a differentiable remedy; the soft clamp addresses this directly.

We note that WW-NTBN partly sacrifices the model-agnostic character of deep hedging. By incorporating the Black-Scholes delta and gamma as structural inputs, the architecture implicitly assumes log-normal dynamics for the underlying asset. This is a deliberate trade-off: in exchange for this additional model assumption, WW-NTBN achieves faster convergence, lower entropic risk at convergence, and no-transaction bands that quantitatively agree with the stochastic control solution across a range of transaction cost levels. In settings where the log-normal assumption is inappropriate, NTBN- Δ remains the better choice.

Comprehensive numerical comparison. We conduct a systematic comparison of the Black-Scholes benchmark, the stochastic control dynamic programming solution, the

deep hedging MLP baseline, NTBN- Δ , and WW-NTBN under identical parameters and across a range of transaction cost levels. We compare writer and buyer indifference prices, no-transaction band geometry, terminal P&L distributions, and trade metrics. We further apply all methods to the bull call spread, documenting the divergence between joint and naive hedging strategies and its implications for pricing.

1.4 Organisation

Section 2 develops the stochastic control framework: we introduce the optimisation problem, derive the HJBQVI, characterise the no-transaction band, and present the Whalley–Wilmott asymptotic formula together with the numerical dynamic programming scheme. Section 3 treats the bull call spread as an application, establishing the non-linearity of prices and hedging strategies under transaction costs. Section 4 introduces the deep hedging framework and describes the three network architectures: the MLP baseline, NTBN- Δ , and WW-NTBN. Section 5 presents the numerical comparison. Section 6 concludes. The Appendix collects the omitted proofs.

Notation. Throughout, S_t denotes the price of the risky asset at time t , W_t the investor’s wealth, y_t the number of shares held, and $\lambda > 0$ the proportional transaction cost rate, assumed symmetric. We write σ for volatility, μ for drift, r for the risk-free rate, K for the option strike, and T for maturity. The risk-aversion parameter of the CARA utility $u(x) = 1 - e^{-\gamma x}$ is $\gamma > 0$. We write Φ for the standard normal CDF, ϕ for its density, and Δ_{BS} , Γ_{BS} for the Black-Scholes delta and gamma of the option under consideration.

2 Optimal Hedging via Stochastic Control

2.1 Market Model and the Frictionless Problem

2.1.1 Price Dynamics and Portfolio Construction

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$ supporting a standard Brownian motion Z_t . The market consists of two assets. The first is a risky asset whose price S_t follows a geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dZ_t, \tag{3}$$

where $\mu \in \mathbb{R}$ is the drift and $\sigma > 0$ the volatility. The second is a risk-free bond with dynamics $dB_t = rB_t dt$, so that one unit of bond at time t is worth $e^{r(T-t)}$ at maturity.

The investor constructs a portfolio by holding y_t shares of the risky asset and allocating the remainder of wealth to the bond. We denote by $W_t = y_t S_t + B_t$ the total portfolio wealth at time t , so the bond position has value $W_t - y_t S_t$. We assume the portfolio is

self-financing: no external cash flows occur, and any purchase of shares is funded by selling bonds and vice versa. The change in wealth over an infinitesimal interval dt is then

$$dW_t = y_t dS_t + (W_t - y_t S_t) \frac{dB_t}{B_t} = y_t dS_t + r(W_t - y_t S_t) dt. \quad (4)$$

Substituting (3) and rearranging,

$$dW_t = rW_t dt + y_t(\mu - r)S_t dt + y_t\sigma S_t dZ_t. \quad (5)$$

It is convenient to write $\pi_t = y_t S_t / W_t$ for the fraction of wealth invested in the risky asset, giving

$$dW_t = [r + \pi_t(\mu - r)]W_t dt + \pi_t\sigma W_t dZ_t. \quad (6)$$

The quantity $\mu - r$ is the excess return of the risky asset over the risk-free rate, often called the risk premium. When $\mu > r$, investing in the risky asset generates a positive expected excess return at the cost of introducing volatility. The investor's problem is to choose π_t to balance this trade-off optimally.

2.1.2 The Optimal Investment Problem and the HJB Equation

The investor has a utility function $u : \mathbb{R} \rightarrow \mathbb{R}$, twice differentiable, strictly increasing, and strictly concave. The goal is to maximise expected utility of terminal wealth,

$$\sup_{\{\pi_t\}_{t \in [0, T]}} \mathbb{E}[u(W_T)]. \quad (7)$$

The optimisation is over adapted processes π_t satisfying suitable integrability conditions. To solve (7) using dynamic programming, we define the *value function* as the maximum achievable expected utility starting from wealth W at time t ,

$$U(t, W) = \sup_{\{\pi_s\}_{s \in [t, T]}} \mathbb{E}_t[u(W_T)], \quad (8)$$

where $\mathbb{E}_t[\cdot]$ denotes expectation conditional on $\mathcal{F}_t$. The value function encodes the full solution to the problem: once U is known, the optimal control at any state (t, W) can be recovered by maximising over π .

By the dynamic programming principle, U satisfies the *Hamilton-Jacobi-Bellman equation*

$$\frac{\partial U}{\partial t} + \max_{\pi} \mathcal{L}^{\pi} U = 0, \quad (9)$$

where $\mathcal{L}^{\pi}$ is the controlled infinitesimal generator of W_t under strategy π . Applying Itô's

formula to $U(t, W_t)$ along the dynamics (6) gives

$$\mathcal{L}^\pi U = \left[r + \pi(\mu - r) \right] W U_W + \frac{1}{2} \pi^2 \sigma^2 W^2 U_{WW}, \quad (10)$$

so the HJB equation reads explicitly

$$U_t + \max_{\pi} \left\{ \left[r + \pi(\mu - r) \right] W U_W + \frac{1}{2} \pi^2 \sigma^2 W^2 U_{WW} \right\} = 0, \quad (11)$$

with terminal condition $U(T, W) = u(W)$. Since $U_{WW} < 0$ by concavity, the maximand is strictly concave in π and the first-order condition yields the *optimal Merton fraction* (Merton, 1969)

$$\pi^* = - \frac{(\mu - r) U_W}{\sigma^2 W U_{WW}}. \quad (12)$$

The quantity $-U_W/(W U_{WW})$ is the reciprocal of the Arrow-Pratt measure of relative risk aversion. Equation (12) thus states that the optimal risky allocation is proportional to the risk premium $\mu - r$ and inversely proportional to risk aversion and variance, a result that holds for any utility function admitting a smooth value function.

2.1.3 Solution under CARA Utility

For the CARA utility $u(W) = 1 - e^{-\gamma W}$, the HJB equation (11) admits an explicit solution. We make the ansatz

$$U(t, W) = 1 - e^{-\gamma \varphi(t) W + \psi(t)}, \quad (13)$$

substitute into (11), and match coefficients. This yields the system $\varphi'(t) + r\varphi(t) = 0$ and $\psi'(t) = (\mu - r)^2/(2\sigma^2)$, with boundary conditions $\varphi(T) = 1$ and $\psi(T) = 0$. Solving gives

$$\varphi(t) = e^{r(T-t)}, \quad \psi(t) = \frac{(\mu - r)^2}{2\sigma^2} (t - T), \quad (14)$$

so that the value function is

$$U(t, W) = 1 - \exp \left(-\gamma e^{r(T-t)} W + \frac{(\mu - r)^2}{2\sigma^2} (t - T) \right). \quad (15)$$

The full derivation is given in Appendix A. Substituting (15) into (12), the optimal fraction and the corresponding optimal number of shares are

$$\pi^* = \frac{\mu - r}{\sigma^2 \gamma W e^{r(T-t)}}, \quad y^* = \frac{\mu - r}{\sigma^2 \gamma S_t e^{r(T-t)}}. \quad (16)$$

Note that y^* is independent of wealth W : under CARA utility, the optimal number of shares held does not depend on how rich the investor is, only on the current stock price and time to maturity. This is a distinctive feature of exponential utility that will prove crucial when we introduce transaction costs.

2.1.4 Adding a Short Option Position

We now suppose the investor is also short one European call option with strike K and maturity T , with payoff $\varphi(S_T) = (S_T - K)^+$. The option is sold at time 0 for a premium p , which is added to the initial wealth. At maturity, the investor must deliver the payoff, so the effective terminal wealth becomes

$$W_T^{\text{option}} = W_T - (S_T - K)^+. \quad (17)$$

The portfolio dynamics (5) are unchanged: the option position does not require any cash flow before maturity, so the self-financing condition still holds. What changes is the terminal condition of the optimisation problem.

Because S_T appears in the terminal wealth (17), the stock price S_t now enters the value function as a state variable. The value function becomes $U(t, S, W)$, and applying Ito's formula now produces cross terms between the S and W dynamics. The HJB equation becomes

$$\frac{\partial U}{\partial t} + \mu S U_S + \frac{1}{2} \sigma^2 S^2 U_{SS} + \max_{\pi} \left\{ [r + \pi(\mu - r)] W U_W + \frac{1}{2} \pi^2 \sigma^2 W^2 U_{WW} + \pi \sigma^2 S W U_{SW} \right\} = 0, \quad (18)$$

with terminal condition $U(T, S, W) = u(W - (S - K)^+)$. The additional term $\pi \sigma^2 S W U_{SW}$ arises because the Brownian motion driving S and W is the same: the investor's wealth and the option's value are exposed to the same source of risk. The first-order condition now gives

$$\pi^* = -\frac{(\mu - r)U_W}{\sigma^2 W U_{WW}} - \frac{S U_{SW}}{W U_{WW}}, \quad (19)$$

which differs from the pure Merton solution (12) by the second term, which reflects the hedging demand induced by the option.

A standard verification argument shows that, under CARA utility, the value function with the option is related to the one without by

$$U^{\text{option}}(t, S, W) = U(t, W - f(t, S)), \quad (20)$$

where $f(t, S)$ is the Black-Scholes price of the call (Davis et al., 1993). In other words, the effect of the short option position is simply to reduce the effective wealth by the option's Black-Scholes value. This is the pricing-hedging duality of the frictionless case: the fair price of the option is exactly $f(t, S)$.

Using (20), we can compute U_{SW} and U_{WW} explicitly in terms of U and the Black-Scholes sensitivities. The key step is that $\partial \tilde{W} / \partial S = -\Delta_{BS}$ where $\tilde{W} = W - f(t, S)$, which gives $U_{SW} = -\Delta_{BS} \cdot U_{WW}$. Substituting into (19) and using the explicit CARA value

function, the optimal number of shares held is

$$y^* = \Delta_{\text{BS}}(t, S) + \frac{\mu - r}{\sigma^2 \gamma S e^{r(T-t)}}, \quad (21)$$

where $\Delta_{\text{BS}} = \partial f / \partial S$ is the Black-Scholes delta of the option. This decomposition has a clean interpretation. The first term, Δ_{BS} , is the standard delta hedge: it replicates the option's exposure to moves in S and would be the entire solution in a risk-neutral world. The second term, $(\mu - r) / (\sigma^2 \gamma S e^{r(T-t)})$, is exactly the Merton speculative position the investor would hold in the absence of any option (16). The two components are fully additive: under CARA utility, the speculative demand depends only on the stock price and time to maturity, not on the investor's wealth or other positions, so the option introduces a pure hedging demand on top of which the investor adds the same speculation as without the option.

When risk aversion $\gamma \rightarrow \infty$, the speculative term vanishes and the investor holds the pure delta hedge $y^* \rightarrow \Delta_{\text{BS}}$: an infinitely risk-averse investor does not speculate and hedges the option perfectly. As $\gamma \rightarrow 0$ the investor becomes risk-neutral and the speculative position dominates, with the delta hedge becoming relatively small.

With the frictionless case fully characterised, we now introduce transaction costs, which fundamentally change the structure of the optimal strategy.

2.2 Transaction Costs: Problem Formulation

We now introduce proportional transaction costs at rate $\lambda > 0$, symmetric for buying and selling. In this section we decompose the portfolio into two separate accounts: y_t shares of the risky asset and a cash balance X_t , so that total wealth is

$$W_t = X_t + y_t S_t. \quad (22)$$

Unlike the frictionless case, trading is no longer costless and the portfolio is not self-financing: every transaction reduces total wealth by λS_t per unit traded.

Trading is governed by two non-negative adapted processes l_t (rate of purchase) and m_t (rate of sale), giving

$$dy_t = l_t dt - m_t dt, \quad (23)$$

$$dX_t = rX_t dt - (1 + \lambda)S_t l_t dt + (1 - \lambda)S_t m_t dt. \quad (24)$$

Each purchase of $l_t dt$ shares costs $(1 + \lambda)S_t l_t dt$ from the cash account (the factor $1 + \lambda$ reflects the transaction cost), while each sale of $m_t dt$ shares generates $(1 - \lambda)S_t m_t dt$ in cash.

Remark 2.1. *To verify consistency with the frictionless case, set $\lambda = 0$ in (23)–(24).*

Total wealth $W_t = X_t + y_t S_t$ then satisfies

$$\begin{aligned}
dW_t &= dX_t + y_t dS_t + S_t dy_t \\
&= \left(rX_t dt - S_t dy_t \right) + y_t (\mu S_t dt + \sigma S_t dZ_t) + S_t dy_t \\
&= r(X_t + y_t S_t) dt + y_t S_t (\mu - r) dt + y_t \sigma S_t dZ_t \\
&= rW_t dt + \pi_t (\mu - r) W_t dt + \pi_t \sigma W_t dZ_t,
\end{aligned} \tag{25}$$

where $\pi_t = y_t S_t / W_t$. This recovers (6) exactly, confirming that the transaction cost formulation is a consistent extension of the frictionless one.

The investor is short a European option with payoff $\varphi(S_T)$. The optimisation problem is

$$\sup_{(l,m) \geq 0} \mathbb{E} \left[u \left(X_T + y_T S_T - \lambda S_T |y_T| - \varphi(S_T) \right) \right], \tag{26}$$

where the liquidation term $-\lambda S_T |y_T|$ accounts for the cost of unwinding the remaining position at maturity. The associated value function is

$$U(t, S, y, X) = \sup_{(l,m) \geq 0} \mathbb{E}_{t,S,y,X} \left[u \left(X_T + y_T S_T - \lambda S_T |y_T| - \varphi(S_T) \right) \right]. \tag{27}$$

2.3 HJB Quasi-Variational Inequality and the No-Transaction Band

Applying Itô's formula to $U(t, S_t, y_t, X_t)$ along the dynamics (23)–(24), and noting that dy_t and the controlled part of dX_t are pure finite-variation processes, the only second-order term is $(dS_t)^2 = \sigma^2 S^2 dt$. Taking expectations, the drift of U must vanish at the optimum by the dynamic programming principle, giving

$$U_t + \mu S U_S + \frac{1}{2} \sigma^2 S^2 U_{SS} + r X U_X + l \left(U_y - (1 + \lambda) S U_X \right) + m \left(-U_y + (1 - \lambda) S U_X \right) = 0. \tag{28}$$

The controls $l, m \geq 0$ appear linearly in (28). Since they are unconstrained in magnitude, the only way (28) can hold without driving the drift to $\pm\infty$ is if each coefficient is non-positive whenever the corresponding control is active. This forces the problem into a quasi-variational inequality structure. Define

$$A = U_y - (1 + \lambda) S U_X, \tag{29}$$

$$B = -U_y + (1 - \lambda) S U_X, \tag{30}$$

$$C = -U_t - \mu S U_S - \frac{1}{2} \sigma^2 S^2 U_{SS} - r X U_X. \tag{31}$$

The Hamilton-Jacobi-Bellman equation then takes the form of the quasi-variational inequality

$$\max\{A, B, C\} = 0, \quad (32)$$

with terminal condition

$$U(T, S, y, X) = u(X + yS - \lambda S|y| - \varphi(S)). \quad (33)$$

Definition 2.2 (Trading regions). *The state space (S, y, X) is partitioned into three regions according to which term in (32) is active.*

- No-transaction region $\mathcal{NT}$: $C = 0$ and $A, B \leq 0$. No trading occurs and U satisfies the PDE $U_t + \mu S U_S + \frac{1}{2} \sigma^2 S^2 U_{SS} + r X U_X = 0$. The condition $A, B \leq 0$ is equivalent to

$$(1 - \lambda)S \leq \frac{U_y}{U_X} \leq (1 + \lambda)S. \quad (34)$$

- Buy region $\mathcal{B}$: $A = 0$ and $B, C \leq 0$. The marginal value of a share exceeds the ask price ($U_y/U_X > (1 + \lambda)S$), so the investor purchases shares continuously, maintaining $U_y/U_X = (1 + \lambda)S$, until re-entering $\mathcal{NT}$ from below.
- Sell region $\mathcal{S}$: $B = 0$ and $A, C \leq 0$. The marginal value of a share falls below the bid price ($U_y/U_X < (1 - \lambda)S$), so the investor sells shares continuously, maintaining $U_y/U_X = (1 - \lambda)S$, until re-entering $\mathcal{NT}$ from above.

The ratio U_y/U_X is the marginal value of an additional share expressed in units of cash. The no-transaction condition (34) states that trading is suboptimal precisely when this marginal value lies within the bid-ask spread $[(1 - \lambda)S, (1 + \lambda)S]$: the utility gain from rebalancing is smaller than the cost it incurs. When the marginal value of a share exceeds the ask price $(1 + \lambda)S$, the investor holds too few shares and buys; when it falls below the bid price $(1 - \lambda)S$, the investor holds too many and sells. Simultaneous buying and selling ($l > 0$ and $m > 0$) is never optimal, as it would waste transaction costs without improving the portfolio.

2.4 CARA Utility and Dimension Reduction

For general utility functions, the value function U depends on all four variables (t, S, y, X) , making direct numerical solution costly. The CARA utility $u(x) = 1 - e^{-\gamma x}$ admits a factorisation that eliminates the cash dimension entirely.

Proposition 2.3 (Dimension reduction). *Under CARA utility, the value function takes the form*

$$U(t, S, y, X) = 1 - e^{-\gamma X/\delta(t,T)} Q(t, y, S), \quad (35)$$

where $\delta(t, T) = e^{-r(T-t)}$ is the discount factor and $Q(t, y, S)$ solves the reduced quasi-variational inequality

$$\min \left\{ Q_y + \frac{\gamma(1+\lambda)S}{\delta(t, T)}Q, -Q_y - \frac{\gamma(1-\lambda)S}{\delta(t, T)}Q, -Q_t - \mu SQ_S - \frac{1}{2}\sigma^2 S^2 Q_{SS} \right\} = 0, \quad (36)$$

with terminal condition $Q(T, y, S) = \exp(-\gamma(yS - \lambda S|y| - \varphi(S)))$.

Proof. See Appendix A. □

The key consequence is that Q does not depend on X . Once Q is computed by solving (36) on the three-dimensional grid (t, y, S) , the full value function U is recovered from (35) for any initial cash balance X at negligible additional cost.

2.5 Indifference Pricing

Utility indifference pricing asks: what cash amount p makes the investor exactly indifferent between selling the option and receiving p , versus not selling? Formally, the writer's indifference price p^w satisfies

$$U^{\text{with}}(t, S, y, X + p^w) = U^{\text{without}}(t, S, y, X), \quad (37)$$

where the superscripts refer to the value functions with and without the option liability respectively. Using the factorisation (35), the cash dependence cancels and one obtains the explicit formula

$$p^w(t, y, S) = \frac{\delta(t, T)}{\gamma} \left[\log Q^{\text{without}}(t, y, S) - \log Q^{\text{with, writer}}(t, y, S) \right]. \quad (38)$$

The buyer's indifference price p^b is defined symmetrically by

$$p^b(t, y, S) = \frac{\delta(t, T)}{\gamma} \left[\log Q^{\text{with, buyer}}(t, y, S) - \log Q^{\text{without}}(t, y, S) \right]. \quad (39)$$

In the presence of transaction costs, $p^w \geq p^b$ in general, generating a bid-ask spread that widens with both λ and γ . This spread vanishes as $\lambda \rightarrow 0$, where both prices converge to the Black-Scholes price.

Remark 2.4. *The factorisation (35) and the resulting pricing formulas (38)–(39) hold because CARA utility is translation-invariant: $u(x + c) = e^{-\gamma c}u(x)$ up to a constant. This is the key analytic advantage of exponential utility in the presence of transaction costs.*

2.6 The Whalley-Wilmott Asymptotic Band

The quasi-variational inequality (36) must in general be solved numerically. However, in the limit of small transaction costs $\lambda \rightarrow 0$, Whalley and Wilmott (1997) show that the width of the no-transaction band can be obtained analytically by a matched asymptotic expansion. The key insight is that when λ is small, the no-transaction band is a narrow region of width $O(\lambda^{1/3})$ around the frictionless optimal holding $y^*(t, S)$. This fractional power arises naturally from the structure of the problem: the suboptimality cost of staying within the band is quadratic in the deviation from y^* , while the transaction cost of trading is linear in the trade size, and balancing these two contributions leads to a cubic optimisation whose minimiser scales as $\lambda^{1/3}$. A heuristic sketch of the derivation is given in Appendix B.

The asymptotic formula. To leading order in λ , the no-transaction band around the frictionless optimal holding $y^*(t, S)$ has half-width

$$h_{\text{WW}}(t, S) = \left(\frac{3\lambda\delta(t, T)S\Gamma_{\text{BS}}(t, S)^2}{2\gamma} \right)^{1/3}, \quad (40)$$

where $\delta(t, T) = e^{-r(T-t)}$ is the discount factor and $\Gamma_{\text{BS}}(t, S) = \partial^2 f / \partial S^2$ is the Black-Scholes gamma of the option in price space. The optimal policy is therefore to do nothing when $y_t \in [y^* - h_{\text{WW}}, y^* + h_{\text{WW}}]$, and otherwise to trade to the nearest boundary.

Interpretation. The formula (40) has clear intuitive content. The half-width h_{WW} increases with λ (wider band when trading is more costly) and with Γ_{BS} (wider band when the option delta changes rapidly with S , making rebalancing errors more persistent and thus more costly to correct). It decreases with γ (tighter band for more risk-averse investors who tolerate less deviation from the optimal position). The $\lambda^{1/3}$ scaling is a well-known feature of singular control problems with proportional costs (Whalley and Wilmott, 1997).

Validity. The Whalley-Wilmott formula is an asymptotic result valid for small λ . For large transaction costs it underestimates the true band width, as confirmed by our numerical experiments in Section 5. Nevertheless, it provides an accurate and computationally free approximation in the small to moderate cost regime ($\lambda \lesssim 1\%$) that is most relevant in practice.

2.7 Numerical Scheme

We solve the reduced quasi-variational inequality (36) by backward induction on a binomial tree for S combined with a uniform grid for y . The CARA dimension reduction of

Proposition 2.3 is essential here: because Q does not depend on X , the scheme operates on a three-dimensional grid (t, y, S) rather than the full four-dimensional state space, making the computation tractable.

Discretisation. We discretise time into N steps of size $\Delta t = T/N$. The stock price follows a binomial approximation with up and down factors

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad (41)$$

and real-world probability $p = (e^{\mu\Delta t} - d)/(u - d)$. The holding y is discretised on a uniform grid $y_k = k\Delta y$ for $k \in \{-M, \dots, M\}$, where $M = \text{int}(0.8\lfloor N/2 \rfloor)$, and $\Delta y = \sigma\Delta t$. We denote by $Q(t, S_j, y_k)$ the approximate value of Q at node (t, S_j, y_k) .

Backward recursion. Starting from the terminal condition, three actions are evaluated at each node: do nothing, buy Δy shares, or sell Δy shares. The exponential factors in the buy and sell updates arise from the CARA factorisation (35): a cash outflow of $(1 + \lambda)S\Delta y$ at time t multiplies Q by $\exp(-\gamma(1 + \lambda)S\Delta y/\delta(t, T))$, since Q absorbs the discounted trading costs. The optimal value is the minimum of the three options, since $Q > 0$ and $U = 1 - e^{-\gamma X/\delta}Q$ is increasing in $-Q$. The full procedure is given in Algorithm 1.

Algorithm 1 Dynamic programming scheme for $Q(t, y, S)$

Require: Grid parameters $N, M, \Delta y$, option payoff φ , model parameters $\mu, \sigma, r, \lambda, \gamma$

- 1: Compute binomial tree nodes S_j and probability p via (41)
 - 2: Initialise holding grid $y_k = k\Delta y, k = -M, \dots, M$
 - 3: **for** each terminal node (S_j, y_k) **do**
 - 4: $Q(T, S_j, y_k) \leftarrow \exp(-\gamma(y_k S_j - \lambda S_j |y_k| - \varphi(S_j)))$
 - 5: **end for**
 - 6: **for** $n = N - 1, \dots, 0$ **do**
 - 7: **for** each node (S_j, y_k) at time $t = n\Delta t$ **do**
 - 8: $Q^{\text{NT}} \leftarrow p Q(t + \Delta t, S_j u, y_k) + (1 - p) Q(t + \Delta t, S_j d, y_k)$
 - 9: $Q^{\text{Buy}} \leftarrow \exp\left(\frac{-\gamma(1 + \lambda)S_j \Delta y}{\delta(t, T)}\right) Q(t, S_j, y_{k+1})$
 - 10: $Q^{\text{Sell}} \leftarrow \exp\left(\frac{\gamma(1 - \lambda)S_j \Delta y}{\delta(t, T)}\right) Q(t, S_j, y_{k-1})$
 - 11: $Q(t, S_j, y_k) \leftarrow \min\{Q^{\text{NT}}, Q^{\text{Buy}}, Q^{\text{Sell}}\}$
 - 12: Record optimal action in $\mathcal{A}(t, S_j, y_k) \in \{\text{NT}, \text{Buy}, \text{Sell}\}$
 - 13: **end for**
 - 14: **end for**
 - 15: Run scheme twice to obtain Q^{with} and Q^{without} (with and without φ)
 - 16: $\hat{p}^w \leftarrow \frac{\delta(0, T)}{\gamma} \left[\log Q^{\text{without}}(0, S_0, y_0) - \log Q^{\text{with}}(0, S_0, y_0) \right]$
 - 17: **return** writer price $\hat{p}^w$, action map $\mathcal{A}$
-

No-transaction band extraction. The action map $\mathcal{A}$ recorded during the backward pass gives the full partition of the (S, y) space into the three regions of Definition 2.2. For each stock price S_j and time t , the no-transaction band boundaries are extracted as

$$\ell^{\text{SC}}(t, S_j) = \min\{y_k : \mathcal{A}(t, S_j, y_k) = \text{NT}\}, \quad u^{\text{SC}}(t, S_j) = \max\{y_k : \mathcal{A}(t, S_j, y_k) = \text{NT}\}. \quad (42)$$

These boundaries are used in Section 5 to compare the stochastic control solution directly with the no-transaction bands learned by NTBN- Δ and WW-NTBN.

Buyer's price. The buyer's indifference price is computed by modifying the terminal condition to reflect the buyer's position: the buyer receives the payoff $\varphi(S_T)$ at maturity, so the terminal condition becomes $Q^{\text{buyer}}(T, S_j, y_k) = \exp(-\gamma(y_k S_j - \lambda S_j |y_k| + \varphi(S_j)))$. The buyer's price then follows from (39) analogously to the writer's price.

3 The Bull Call Spread under Transaction Costs

3.1 Definition and the Frictionless Case

A bull call spread consists of a long position in a European call with strike K_1 and a short position in a European call with the same maturity T and a higher strike $K_2 > K_1$, written on the same underlying. The terminal payoff is

$$\varphi_{\text{CS}}(S_T) = (S_T - K_1)^+ - (S_T - K_2)^+. \quad (43)$$

This strategy allows the investor to benefit from a moderate rise in the underlying while paying a lower premium than for a single call, since the short leg partially offsets the cost of the long leg.

In the frictionless case, the pricing problem is linear. The Black-Scholes PDE is linear in the option value, so the price of any combination of options is simply the corresponding combination of individual prices. For the bull call spread this gives

$$p_{\text{CS}} = f(t, S; K_1) - f(t, S; K_2), \quad (44)$$

where $f(t, S; K)$ denotes the Black-Scholes price of a call with strike K . Equivalently, using the duality relation (20), the value function of an investor short the spread satisfies

$$U^{\text{CS}}(t, S, W) = U\left(t, W - f(t, S; K_1) + f(t, S; K_2)\right), \quad (45)$$

and the optimal hedge is $\Delta_{\text{BS}}(t, S; K_1) - \Delta_{\text{BS}}(t, S; K_2)$, the difference of the two individual deltas. Linearity of both pricing and hedging is a direct consequence of the linearity of

the frictionless HJB equation.

3.2 Breakdown of Linearity under Transaction Costs

Once proportional transaction costs are introduced, the linearity argument breaks down at a precise point. Under transaction costs, pricing and hedging are governed by the quasi-variational inequality (32), which is nonlinear. The key object that determines the structure of the optimal strategy is the no-transaction band, whose half-width is given to leading order by the Whalley-Wilmott formula (40):

$$h_{\text{WW}}(t, S) = \left(\frac{3\lambda\delta(t, T)S\Gamma^2(t, S)}{2\gamma} \right)^{1/3}, \quad (46)$$

where $\Gamma(t, S)$ is the gamma of the option being hedged. The width of the no-transaction band is therefore a nonlinear function of the gamma, scaling as $\Gamma^{2/3}$.

Now consider hedging the two legs of the call spread separately. Each leg has its own no-transaction band, with half-widths $h_{\text{WW}}(t, S; K_1)$ and $h_{\text{WW}}(t, S; K_2)$ respectively. When the two strategies are combined, a rebalancing trade is triggered whenever the joint holding exits either individual band. The effective no-transaction band for the combined position is therefore the intersection of the two individual bands, which is generically much narrower than either band alone. Near the money, where the gammas of the two calls are largest and most similar, this intersection can nearly vanish, forcing almost continuous rebalancing and generating transaction costs that approach those of delta-hedging without any band structure at all.

The correct approach is to treat the call spread as a single instrument and apply the QVI framework directly to the combined payoff (43). The gamma of the spread is

$$\Gamma_{\text{CS}}(t, S) = \Gamma_{\text{BS}}(t, S; K_1) - \Gamma_{\text{BS}}(t, S; K_2), \quad (47)$$

which is strictly smaller in magnitude than either individual gamma when the two calls are close to the money, since the two gammas partially cancel. By the Whalley-Wilmott formula, this smaller gamma yields a wider no-transaction band for the jointly hedged spread than the intersection of the two individual bands. The joint hedger therefore trades less frequently, incurs lower transaction costs, and consequently prices the spread at a lower indifference price than the naive approach.

This non-linearity has a direct consequence for pricing. Define the naive spread price as the difference of the two individual indifference prices,

$$p_{\text{naive}} = p^w(K_1) - p^b(K_2), \quad (48)$$

where $p^w(K_1)$ is the writer's indifference price for the K_1 call and $p^b(K_2)$ is the buyer's

indifference price for the K_2 call. The joint indifference price p_{CS} satisfies

$$p_{\text{CS}} \leq p_{\text{naive}}, \quad (49)$$

with equality only when $\lambda = 0$. The gap $p_{\text{naive}} - p_{\text{CS}}$ grows with the transaction cost level λ and reflects the hedging inefficiency of treating the two legs independently. This is confirmed numerically in Section 5.

Remark 3.1. *The argument above illustrates a general principle: under transaction costs, the price of a portfolio of derivatives is not the sum of the individual prices. The indifference pricing functional is subadditive for a writer, meaning that hedging multiple liabilities jointly is always at least as efficient as hedging them separately. This is precisely because the no-transaction bands of the joint problem are wider than the intersection of the individual bands, reducing the total cost of hedging.*

3.3 Numerical Treatment

The joint hedging problem for the call spread fits directly into the framework of Section 2. It suffices to replace the terminal condition (33) with

$$U(T, S, y, X) = u(X + yS - \lambda S|y| - \varphi_{\text{CS}}(S)), \quad (50)$$

where φ_{CS} is given by (43), and run the dynamic programming scheme of Section 2.7 unchanged. The indifference price is then computed via (38) by comparing the value function with and without the spread liability.

For the naive approach, the two legs are priced and hedged independently using the same scheme, and the resulting strategies are superimposed. The comparison of the two approaches in terms of no-transaction band geometry, terminal P&L distributions, trade metrics, and prices is presented in Section 5.

4 Deep Hedging

4.1 From Stochastic Control to Neural Networks

The stochastic control framework developed in Section 2 provides a rigorous and interpretable solution to the hedging problem under transaction costs. However, it faces fundamental limitations as a practical tool. The dynamic programming scheme requires discretising a state space of dimension four (t, S, y, X) , which becomes computationally prohibitive as the grid is refined. The approach also requires strong modelling assumptions: the derivation of the HJB equation, the dimension reduction of Proposition 2.3, and the Whalley-Wilmott approximation all rely on the specific structure of the GBM dynamics,

and extending the framework to richer price processes typically destroys the tractability that makes it appealing. Finally, the scheme scales poorly to portfolios of multiple derivatives, where the state space dimension grows with the number of instruments.

Deep hedging (Buehler et al., 2019) addresses these limitations by replacing the explicit solution of the dynamic programming equations with a direct optimisation over the parameters of a neural network. The network learns a hedging policy from simulated market trajectories, and the computational cost scales with the number of paths rather than the dimension of the state space, making the approach naturally suited to complex payoffs and multi-instrument portfolios. The MLP and NTBN- Δ architectures we consider are fully model-agnostic: they require only the ability to simulate paths of the underlying, with no assumption on the price dynamics. The WW-NTBN uses the Black-Scholes gamma as a structural prior, which introduces a dependence on GBM dynamics, but this prior could in principle be replaced by a model-specific sensitivity under richer dynamics, preserving the same architectural advantages. As we will show, the deep hedging approach produces hedging strategies and indifference prices that are quantitatively consistent with the stochastic control solution in the GBM setting, providing a valuable cross-validation of both approaches.

4.2 Problem Formulation and Loss Function

We retain the same economic objective as in Section 2: the investor seeks to minimise risk when holding a short position in a European option with payoff $\varphi(S_T)$, subject to proportional transaction costs at rate λ . For simplicity we set $r = 0$ throughout this section, so discounting plays no role.

Rather than maximising expected CARA utility directly, we minimise the *entropic risk measure*

$$\rho_\gamma(X) = \frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma X}], \quad (51)$$

where X is the terminal P&L of the hedging strategy and $\gamma > 0$ is the risk-aversion parameter. Minimising $\rho_\gamma(X)$ is equivalent to maximising $1 - \mathbb{E}[-e^{-\gamma X}]$, which is the objective associated with the CARA utility $u(x) = 1 - e^{-\gamma x}$. The entropic risk measure is preferred numerically because it is a convex risk measure (Föllmer and Schied, 2002), yielding a well-behaved optimisation landscape with stable gradients.

The terminal P&L for a hedging strategy $\{y_t\}_{t \leq T}$ is

$$X = -\varphi(S_T) + \sum_{t=0}^{T-\Delta t} y_t(S_{t+\Delta t} - S_t) - \sum_{t=0}^{T-\Delta t} \lambda S_t |y_t - y_{t-1}|, \quad (52)$$

where the first term is the option liability, the second is the cumulative gain from the delta position, and the third is the total transaction cost incurred. The hedge is initialised at $y_{-1} = 0$.

From optimal control to neural network parametrisation. In the stochastic control formulation of Section 2, the investor directly controls the trading rate (l_t, m_t) at each time t , and the optimal policy is a function of the full state (t, S_t, y_t, X_t) . In the deep hedging framework, we instead seek the optimal policy within the class of functions representable by a neural network. Formally, let $I_t \in \mathbb{R}^d$ denote the feature vector available to the investor at time t (for instance, log-moneyness, time to maturity, and volatility). A neural network with parameters θ defines a deterministic policy

$$y_t = \pi_\theta(I_t, y_{t-1}), \quad (53)$$

mapping the current market information and previous position to the new hedge. The original optimisation over adapted processes $\{y_t\}$ is thereby replaced by a finite-dimensional optimisation over θ :

$$\min_{\theta} \rho_\gamma(X^\theta) = \min_{\theta} \frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma X^\theta}], \quad (54)$$

where X^θ denotes the terminal P&L (52) under the policy π_θ . This is a standard stochastic optimisation problem that can be solved by gradient-based methods: given a batch of simulated paths, the loss $\rho_\gamma(X^\theta)$ is evaluated, its gradient with respect to θ is computed by backpropagation through the sequence of hedge decisions (53), and θ is updated using the Adam optimiser (Kingma and Ba, 2015).

The expressiveness of the policy class depends on the network architecture. A sufficiently deep and wide network can approximate any measurable function of I_t (Hornik, 1991), so in principle the neural network policy can converge to the true optimal policy as the network size grows. In practice, the choice of architecture matters greatly: a poorly designed architecture may require many more parameters and training samples to learn the same policy. The three architectures we consider in Section 4.3 differ precisely in how much structural knowledge of the optimal policy is embedded in the network design.

Neural network structure. A feedforward neural network with L hidden layers computes

$$x^{(0)} = I_t, \quad x^{(\ell+1)} = \sigma(W^{(\ell)}x^{(\ell)} + b^{(\ell)}), \quad \ell = 0, \dots, L-1, \quad (55)$$

where $W^{(\ell)}$ and $b^{(\ell)}$ are the weight matrix and bias vector of layer ℓ , and σ is a nonlinear activation function applied elementwise. The output layer is linear and produces a scalar or vector depending on the architecture. The activation function introduces the nonlinearity necessary to approximate complex input-output mappings; without it, the entire network would collapse to a single affine map regardless of depth. We use the ReLU activation $\sigma(x) = \max(0, x)$ throughout, which preserves gradient flow during training while being computationally efficient. The full parameter set is $\theta = \{W^{(\ell)}, b^{(\ell)}\}_{\ell=0}^L$, optimised end-to-end by minimising (54).

4.3 Network Architectures

At each rebalancing date t , the network receives a feature vector $I_t \in \mathbb{R}^d$ and outputs a hedging decision. The feature vector consists of the log-moneyness $\log(S_t/K)$, the time to maturity $T - t$, and the volatility σ . All three architectures share this input structure; they differ in how they map I_t to the hedge y_t .

4.3.1 MLP Baseline

The first architecture is a standard feedforward Multi-Layer Perceptron (MLP). At each time step, the network takes as input the feature vector I_t augmented with the previous hedge y_{t-1} , and outputs the new hedge directly:

$$y_t = \text{MLP}_\theta(I_t, y_{t-1}). \quad (56)$$

The network consists of five hidden layers with 32 neurons each and ReLU activations, followed by a linear output layer producing a scalar. The inclusion of y_{t-1} as an input allows the network to account for the transaction cost of moving from the current position, but introduces path-dependence into the optimisation landscape: the network must simultaneously learn the target hedge level and the cost of reaching it from the current position. This action-dependence can slow convergence and introduce instability, particularly at high transaction cost levels. Despite this limitation, the MLP serves as a useful baseline and provides insight into the behaviour of unconstrained learned strategies.

4.3.2 NTBN- Δ

The No-Transaction Band Network of [Imaki et al. \(2023\)](#) addresses the action-dependence of the MLP by explicitly modelling the no-transaction band structure known from stochastic control theory. Rather than outputting a hedge directly, the network outputs the lower and upper boundaries (ℓ_t, u_t) of a no-transaction band, and the hedge is updated via a clamping operation:

$$y_t = \text{clamp}(y_{t-1}, \ell_t, u_t) = \begin{cases} \ell_t & \text{if } y_{t-1} < \ell_t, \\ y_{t-1} & \text{if } y_{t-1} \in [\ell_t, u_t], \\ u_t & \text{if } y_{t-1} > u_t. \end{cases} \quad (57)$$

This design removes y_{t-1} from the network input: the network receives only the market features I_t and learns when to rebalance rather than how much to rebalance, substantially simplifying the optimisation problem.

In their experimental implementation, [Imaki et al. \(2023\)](#) already centre the band around the Black-Scholes delta: the two network outputs are added and subtracted from

Δ_{BS} to obtain u_t and ℓ_t respectively. However, this choice is not presented as an explicit architectural decision and no theoretical motivation is given for it. We formalise it here as a principled design choice, justified directly by the stochastic control analysis: the frictionless optimum (21) shows that the natural centre of the no-transaction band is the Black-Scholes delta $\Delta_{\text{BS}}(t, S)$, and centring the band there from the outset removes the need for the network to discover this structure from data.

The resulting architecture, which we call *NTBN- Δ* to make the delta-centring explicit, outputs two half-width corrections $(\delta_\ell, \delta_u) \in \mathbb{R}^2$, and the band boundaries are

$$\ell_t = \Delta_{\text{BS}}(t, S_t) - \text{LeakyReLU}(\delta_\ell), \quad u_t = \Delta_{\text{BS}}(t, S_t) + \text{LeakyReLU}(\delta_u), \quad (58)$$

where LeakyReLU ensures the half-widths are non-negative while maintaining gradient flow. The hedge is then computed via (57) without introducing any additional model assumptions beyond those already implicit in the Black-Scholes delta, it serves as the baseline on which WW-NTBN is built.

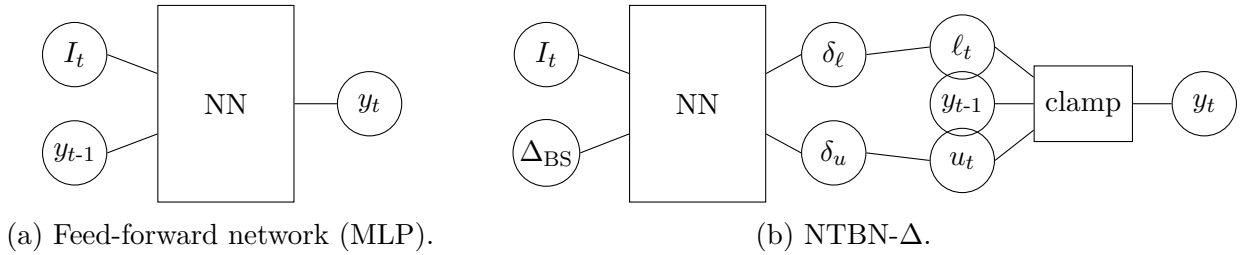


Figure 1: Comparison of the MLP and NTBN- Δ architectures. $\ell_t = \Delta_{\text{BS}} - \text{LeakyReLU}(\delta_\ell)$ and $u_t = \Delta_{\text{BS}} + \text{LeakyReLU}(\delta_u)$.

4.3.3 WW-NTBN

While NTBN- Δ correctly centres the band, it still leaves the bandwidth to be learned entirely from data. The Whalley–Wilmott formula (40) provides an analytical expression for the half-bandwidth that is accurate for small to moderate transaction costs and encodes the correct scaling with λ and γ . We propose to use this as a structural prior, allowing the network to learn only a residual correction on top of the analytical baseline.

The *Whalley–Wilmott No-Transaction Band Network* (WW-NTBN) incorporates two improvements over NTBN- Δ . The first is the WW-guided bandwidth. Given the feature vector $I_t = (\log(S_t/K), T - t, \sigma)$, we compute the Black-Scholes gamma

$$\Gamma_{\text{BS}}(t, S_t) = \frac{\phi(d_1)}{\sigma S_t \sqrt{T - t}}, \quad d_1 = \frac{\log(S_t/K) + \frac{1}{2}\sigma^2(T - t)}{\sigma \sqrt{T - t}}, \quad (59)$$

and the Whalley–Wilmott half-width

$$h_{\text{WW}}(t, S_t) = \left(\frac{3\lambda S_t \Gamma_{\text{BS}}^2(t, S_t)}{2\gamma} \right)^{1/3}. \quad (60)$$

The MLP then outputs two residual corrections $(\epsilon_\ell, \epsilon_u) \in \mathbb{R}^2$, and the band boundaries are

$$\ell_t = \Delta_{\text{BS}}(t, S_t) - \text{LeakyReLU}(h_{\text{WW}}(t, S_t) + \epsilon_\ell), \quad u_t = \Delta_{\text{BS}}(t, S_t) + \text{LeakyReLU}(h_{\text{WW}}(t, S_t) + \epsilon_u). \quad (61)$$

At initialisation, the residual corrections are zero and the band reduces exactly to the Whalley–Wilmott approximation. The network therefore begins training with the correct asymptotic structure and learns corrections to it, rather than learning the band from scratch.

The second improvement is the replacement of the hard clamp (57) with a differentiable *soft clamp*. The hard clamp has zero gradient with respect to y_{t-1} whenever the previous hedge lies outside the band, which removes the learning signal for trajectories where the position is far from the band. We replace it with

$$\text{softclamp}(x, \ell, u) = m + h \cdot \tanh\left(\beta \cdot \frac{x - m}{h}\right), \quad (62)$$

where $m = (\ell + u)/2$ is the band midpoint, $h = (u - \ell)/2$ is the half-width, and $\beta > 0$ controls the sharpness of the transition. For large β , (62) approximates the hard clamp; for moderate β (we use $\beta = 10$ in our experiments), it provides non-zero gradients everywhere, giving the network a learning signal even when the previous hedge is well outside the band. The full forward pass of the WW-NTBN at time t is summarised in Algorithm 2.

Algorithm 2 WW-NTBN forward pass at time t

Require: Features $I_t = (\log(S_t/K), T - t, \sigma)$, previous hedge y_{t-1} , parameters λ, γ, β

- 1: Compute $\Delta_{\text{BS}}(t, S_t)$ and $\Gamma_{\text{BS}}(t, S_t)$ from I_t (closed-form BS formulas)
- 2: $h_{\text{WW}} \leftarrow \left(3\lambda S_t \Gamma_{\text{BS}}^2 / (2\gamma)\right)^{1/3}$ (WW half-width prior)
- 3: $(\epsilon_\ell, \epsilon_u) \leftarrow \text{MLP}_\theta(I_t)$ (residual corrections)
- 4: $\ell_t \leftarrow \Delta_{\text{BS}} - \text{LeakyReLU}(h_{\text{WW}} + \epsilon_\ell)$
- 5: $u_t \leftarrow \Delta_{\text{BS}} + \text{LeakyReLU}(h_{\text{WW}} + \epsilon_u)$
- 6: $y_t \leftarrow \text{softclamp}(y_{t-1}, \ell_t, u_t)$ (differentiable band enforcement)
- 7: **return** y_t

We note that WW-NTBN partly sacrifices the model-agnostic character of deep hedging. By incorporating Δ_{BS} and Γ_{BS} as structural inputs, the architecture implicitly assumes log-normal dynamics for the underlying asset. This is a deliberate trade-off: the WW prior encodes structure that would otherwise require many training samples to discover, yielding faster convergence and better generalisation across transaction cost regimes. In settings

where the log-normal assumption is inappropriate, NTBN- Δ remains the better choice.

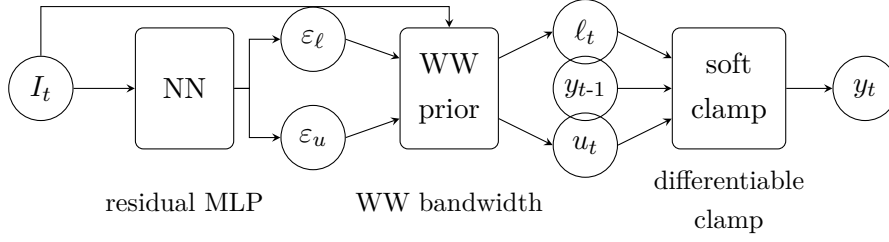


Figure 2: Forward pass of the WW-NTBN.

4.4 Training Algorithm

All three networks are trained using the same procedure, summarised in Algorithm 3. The key steps are the simulation of GBM paths, the application of the network policy to compute terminal P&L, and the minimisation of the entropic risk measure via stochastic gradient descent.

Algorithm 3 Deep hedging training

Require: Network parameters θ , risk aversion γ , transaction cost λ , number of epochs E , batch size N , learning rate η

- 1: **for** $e = 1, \dots, E$ **do**
 - 2: Sample N paths $\{S_t^{(i)}\}_{t=0}^T$ of GBM with parameters (μ, σ)
 - 3: **for** each path $i = 1, \dots, N$ **do**
 - 4: Initialise $y_{-1}^{(i)} = 0$
 - 5: **for** $t = 0, \Delta t, \dots, T - \Delta t$ **do**
 - 6: Compute $I_t^{(i)}$ from $S_t^{(i)}$
 - 7: $y_t^{(i)} \leftarrow \pi_\theta(I_t^{(i)}, y_{t-1}^{(i)})$ (network forward pass)
 - 8: **end for**
 - 9: Compute terminal P&L $X^{(i)}$ via (52)
 - 10: **end for**
 - 11: $\mathcal{L}(\theta) \leftarrow \frac{1}{\gamma} \log\left(\frac{1}{N} \sum_{i=1}^N e^{-\gamma X^{(i)}}\right)$
 - 12: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}(\theta)$ (Adam optimiser)
 - 13: **end for**
 - 14: **return** θ
-

4.5 Indifference Pricing

Once the network is trained, the writer's indifference price is estimated on an independent out-of-sample set of N_{eval} paths. For each path, two terminal P&L values are computed: X^{no} using the trained strategy without the option liability, and X^{with} with the liability

included. The writer’s price is then

$$\hat{p}^w = \frac{1}{\gamma} \log \left(\frac{\frac{1}{N} \sum_i e^{-\gamma X_i^{\text{no}}}}{\frac{1}{N} \sum_i e^{-\gamma X_i^{\text{with}}}} \right), \quad (63)$$

which is the empirical analogue of the theoretical formula (38). The buyer’s price is computed symmetrically. This direct correspondence between the deep hedging price (63) and the stochastic control price (38) is what makes the comparison in Section 5 meaningful: both approaches are computing the same economic quantity, the utility indifference price, by different methods.

5 Numerical Results

5.1 Experimental Setup

All experiments use the following shared parameters: $S_0 = K = 1.0$, $\sigma = 0.2$, $r = \mu = 0$, $\gamma = 1.0$, $T = 1.0$. Transaction costs are tested at five levels: $\lambda \in \{0\%, 0.1\%, 0.5\%, 1\%, 5\%\}$. For the call spread, $K_1 = 0.9$ and $K_2 = 1.1$.

The stochastic control scheme (Algorithm 1) uses $N = 400$ time steps and a holding grid of $2M + 1$ points. We note that the numerical scheme is restricted to trades of exactly $\pm \Delta y$ shares at each step, which is a simplification of the continuous-control problem and introduces discretisation error in the optimal strategy. The deep hedging models are trained using the Adam optimiser (Kingma and Ba, 2015) with a learning rate of 10^{-2} . Following Imaki et al. (2023), who showed that a relatively high learning rate leads to faster and stable convergence for band-based architectures, we use this value uniformly across all three architectures to ensure a fair comparison. All networks consist of five hidden layers with 32 units each and ReLU activations. Prices are evaluated on 10,000 out-of-sample paths via (63). The five models compared throughout are: the Black-Scholes closed-form price (BS), the stochastic control dynamic programming solution (SC), the MLP baseline, NTBN- Δ , and WW-NTBN.

5.2 Frictionless Benchmark

Before examining the effect of transaction costs, we validate all models against the Black-Scholes closed-form price at $\lambda = 0$. In the frictionless case with $\mu = r$, the indifference price under CARA utility coincides with the Black-Scholes price regardless of γ (see Section 2.1), so all models should recover $f_{\text{BS}} = 0.07966$.

Table 1: Option prices at $\lambda = 0$ compared to the Black-Scholes reference.

Model	BS	SC	MLP	NTBN- Δ	WW-NTBN
Price	0.07966	0.07995	0.07983	0.08060	0.07965
Error	–	+0.00029	+0.00017	+0.00094	–0.00001

Table 1 shows that all models are in close agreement with the BS reference. WW-NTBN achieves the closest match with an error of -0.00001 , followed by MLP (0.00017) and SC (0.00029). NTBN- Δ shows the largest deviation (0.00094), which we attribute to the network requiring more training epochs to precisely locate the band width when starting without a WW prior. These results validate the implementations and confirm that all models correctly reduce to the frictionless case when $\lambda = 0$.

5.3 Writer and Buyer Indifference Prices

Table 2: Writer indifference prices for all models across transaction cost levels. The BS reference applies only at $\lambda = 0$.

λ	BS	SC	MLP	NTBN- Δ	WW-NTBN
0%	0.07966	0.07995	0.07983	0.08060	0.07965
0.1%	–	0.08121	0.08174	0.08131	0.08095
0.5%	–	0.08507	0.08426	0.08326	0.08338
1%	–	0.08932	0.08639	0.08597	0.08591
5%	–	0.11750	0.09005	0.08980	0.08878

Table 2 reports writer indifference prices across all transaction cost levels. Several patterns emerge.

First, all models agree that prices increase monotonically with λ , consistent with the theoretical prediction that higher transaction costs raise the minimum compensation required by the writer.

Second, the SC prices increase substantially faster with λ than the DH prices. At $\lambda = 5\%$, SC gives a writer price of 0.11750 while WW-NTBN gives 0.08878 , a gap of nearly 0.03 . This divergence has several contributing factors. On the SC side, the binomial tree discretisation and the restriction to trades of exactly $\pm\Delta y$ per step mean the numerical scheme can only approximate the true optimal continuous-control solution, and this approximation error is likely to grow with λ as the optimal strategy becomes more sensitive to the exact band boundaries. On the DH side, the networks are trained by minimising the empirical entropic risk over a finite batch of simulated paths, which means extreme scenarios that are rare in the training distribution may not be adequately priced.

The SC solution, by considering every node of the binomial tree including those arising from large price movements, produces more conservative prices that account for such scenarios. Finally, the two approaches use fundamentally different time discretisations (a binomial tree for SC and Monte Carlo paths for DH), which can produce systematically different price estimates even when both approximate the same continuous-time problem. We note this divergence as an interesting open question; separating the contribution of each factor would require a more refined numerical study.

Third, among the DH models, WW-NTBN consistently produces the lowest writer prices, followed closely by NTBN- Δ and then MLP. The gap between WW-NTBN and MLP grows with λ : at $\lambda = 5\%$ the difference is 0.00127, suggesting that the WW structural prior becomes increasingly valuable at higher friction levels.

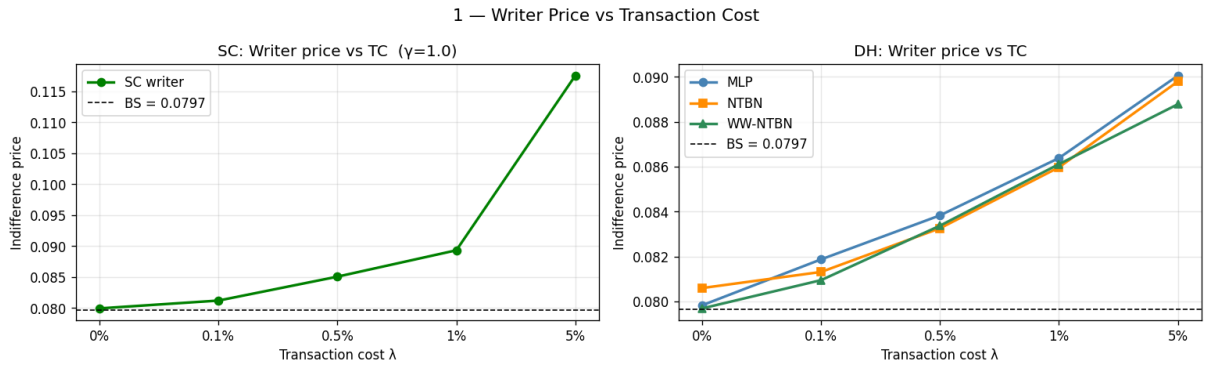


Figure 3: Writer indifference prices vs transaction cost level for all models. The BS reference price at $\lambda = 0$ is shown as a horizontal dashed line.

Table 3: Writer and buyer indifference prices for SC and WW-NTBN, with resulting bid-ask spread.

λ	SC		WW-NTBN		Spread	
	Writer	Buyer	Writer	Buyer	SC	WW-NTBN
0%	0.07995	0.07927	0.07969	0.07905	0.00068	0.00064
0.1%	0.08121	0.07802	0.08095	0.07696	0.00319	0.00399
0.5%	0.08507	0.07427	0.08338	0.07588	0.01080	0.00750
1%	0.08932	0.07037	0.08597	0.07272	0.01895	0.01325
5%	0.11750	0.04129	0.08878	0.07068	0.07621	0.01810

Table 3 and Figure 4 report writer and buyer prices alongside the resulting bid-ask spread for SC and WW-NTBN. A notable feature of the transaction cost framework is the appearance of a bid-ask spread in indifference prices: even under the same model and the same utility function, the writer demands more than the buyer is willing to pay, because the two face different hedging problems. The writer must hedge a short position, the buyer

a long one, and under proportional transaction costs these two hedging programmes have different costs. Both approaches confirm the theoretical prediction of Section 2.5: the spread widens monotonically with λ . The SC spread grows dramatically, reaching 0.07621 at $\lambda = 5\%$, reflecting the conservative nature of the DP solution under high frictions. The WW-NTBN spread grows more moderately, reaching 0.01810 at $\lambda = 5\%$, consistent with the same factors discussed above for the writer prices.

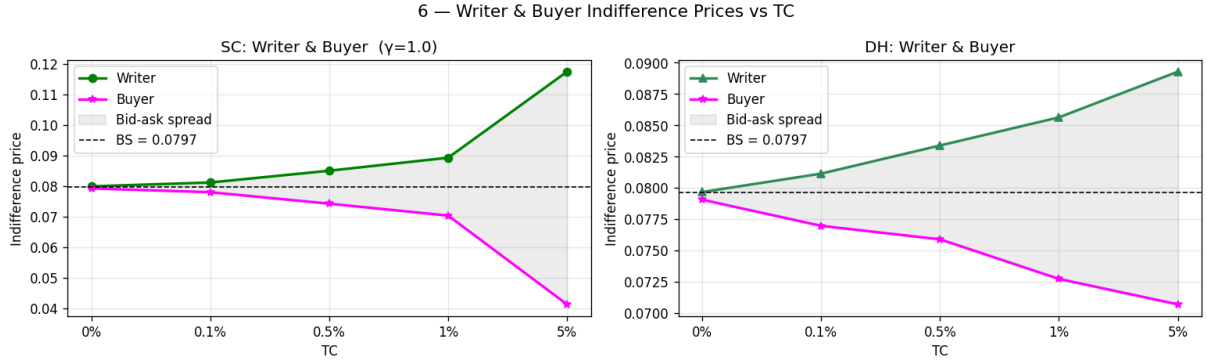


Figure 4: Writer and buyer indifference prices vs transaction cost level for SC (left) and WW-NTBN (right), illustrating the widening bid-ask spread.

5.4 No-Transaction Band Structure

Figure 5 compares the no-transaction band boundaries at $t = 0.1$ extracted from the SC action map via (42) with those learned by NTBN- Δ and WW-NTBN, at $\lambda = 0.1\%$ and $\lambda = 1\%$ respectively. The bands are plotted as a function of log-moneyness $x = \log(S/K)$, with the Black-Scholes delta shown as a dashed reference.

At both TC levels, all models produce bands centred close to Δ_{BS} , consistent with the stochastic control analysis. WW-NTBN produces bands that match the SC boundaries more closely than NTBN- Δ across the full range of log-moneyness, and this advantage holds at both low and high TC. At low TC, the WW prior provides the correct $\lambda^{1/3}$ bandwidth scaling from initialisation, so the residual correction is small and the band rapidly converges to the SC solution. At high TC, where the WW approximation is less exact, the network learns a larger correction but still benefits from the structural initialisation. NTBN- Δ must learn the bandwidth entirely from data at all TC levels, requiring more training and introducing more variance in the learned boundaries.

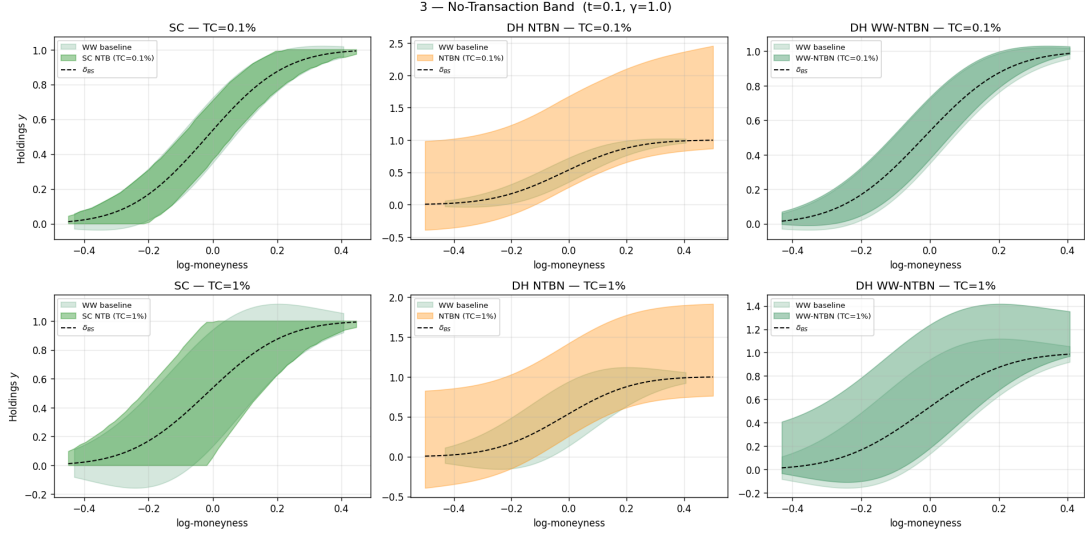


Figure 5: No-transaction band boundaries at $t = 0.1$ for $\lambda = 0.1\%$ and $\lambda = 1\%$.

5.5 Training Convergence

Figure 6 shows the entropic training loss as a function of epoch for all three DH architectures at each TC level. WW-NTBN converges to a lower final loss than both MLP and NTBN- Δ at all TC levels and reaches this loss in fewer epochs. The advantage is most pronounced at low TC ($\lambda \leq 0.5\%$), where the WW prior provides an accurate initialisation.

The MLP consistently converges slowest and to the highest loss, particularly at high TC where action-dependence introduces the most instability. NTBN- Δ lies between the two: it benefits from the delta-centring but must learn the bandwidth from scratch, requiring more epochs than WW-NTBN but fewer than the MLP.



Figure 6: Entropic training loss vs epoch for MLP, NTBN- Δ , and WW-NTBN at each of the five transaction cost levels.

5.6 Terminal P&L Distributions

Figure 7 shows the terminal P&L distributions at $\lambda = 0.5\%$ and $\lambda = 5\%$. All DH models share approximately the same mean P&L by construction of the indifference pricing step. The SC solution achieves a slightly higher mean, reflecting its conservative pricing.

The models differ more meaningfully in tail behaviour. SC exhibits a worse CVaR than all three DH models, meaning its worst-case outcomes are larger in magnitude. Among the DH models, WW-NTBN achieves the best CVaR at both TC levels, followed by NTBN- Δ and then MLP. This ordering is consistent with the band quality results of Section 5.4: a more accurate no-transaction band reduces unnecessary trading and limits the tail losses arising from excess transaction costs.

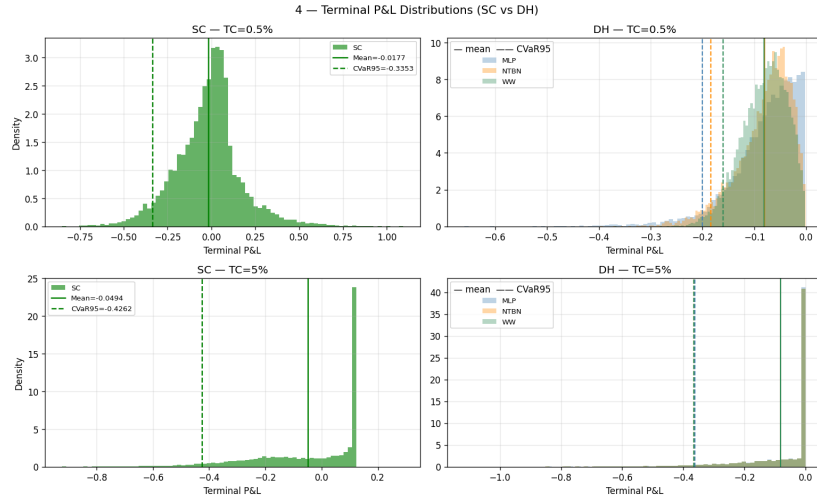


Figure 7: Terminal P&L distributions for SC and DH models at $\lambda = 0.5\%$ and $\lambda = 5\%$.

5.7 Trade Metrics

To quantify the trading behaviour of each model, we introduce two metrics. The *trade frequency* (TF) measures the fraction of nodes at which a rebalancing trade occurs,

$$\text{TF} = \frac{2}{(N+1)(N+2)} \sum_{t=1}^N \sum_{i=0}^{t-1} \sum_{j \in \{i, i+1\}} \mathbf{1}_{|y(t,j) - y(t-1,i)| > 0}, \quad (64)$$

and the *average shares traded* (AS) measures the average absolute position change per node,

$$\text{AS} = \frac{2}{(N+1)(N+2)} \sum_{t=1}^N \sum_{i=0}^{t-1} \sum_{j \in \{i, i+1\}} |y(t,j) - y(t-1,i)|. \quad (65)$$

Both metrics are computed over the binomial tree for SC and over Monte Carlo paths for DH models.

Figure 8 shows both metrics as a function of λ . All models reduce trading frequency and position size monotonically as λ increases, consistent with the widening of the

no-transaction band. The band-based architectures (NTBN- Δ and WW-NTBN) trade significantly less frequently than the MLP at all TC levels. The MLP must learn to reduce trading through the loss function alone, which results in excess trading particularly at higher TC levels.

The SC trade metrics should be interpreted with care. Because the numerical scheme is restricted to trades of exactly $\pm\Delta y$ shares at each node, the SC strategy cannot execute the continuous adjustments that the true optimal policy would make. This restriction means the scheme naturally trades more frequently than a continuous optimal policy would, and the average shares traded is mechanically close to $\Delta y = 0.01$ per trade. Despite this limitation, the qualitative behaviour is correct: both TF and AS decrease monotonically with λ for SC, confirming that the no-transaction band widens as transaction costs increase, even if the absolute levels are influenced by the discretisation.

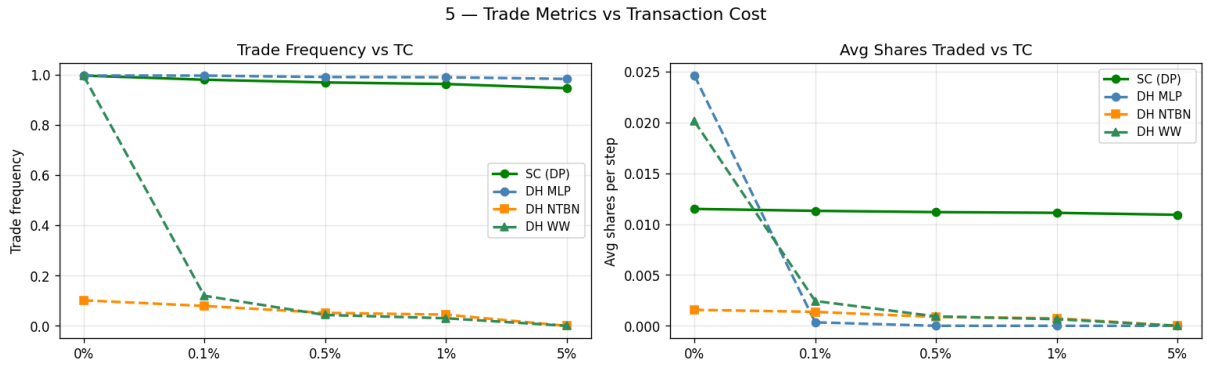


Figure 8: Trade frequency (left) and average shares traded (right) vs transaction cost level for all models.

5.8 Bull Call Spread: Pricing

Table 4: Bull call spread writer prices: joint vs naive hedging for SC and DH (WW-NTBN), with BS reference $p_{BS} = 0.09324$.

λ	SC Joint	SC Naive	DH Joint	DH Naive
0%	0.09324	0.09377	0.09352	0.09358
0.1%	0.09436	0.09400	0.09416	0.09745
0.5%	0.09503	0.09541	0.09524	0.10087
1%	0.09647	0.09733	0.09635	0.10564
5%	0.09677	0.11478	0.09677	0.11163

Table 4 and Figure 9 report joint and naive writer prices for the bull call spread for both SC and DH. At $\lambda = 0$, joint and naive prices are nearly identical for both approaches, consistent with the linearity result of Section 3. As λ increases, the gap between joint

and naive prices widens substantially. At $\lambda = 5\%$, SC naive reaches 0.11478 while SC joint stays at 0.09677, a difference of 0.01801; DH naive reaches 0.11163 while DH joint is 0.09677, a difference of 0.01486. This directly validates the subadditivity argument of Section 3.

The SC joint price grows slowly with λ and appears to approach a ceiling around 0.097. This reflects the structure of the spread payoff: as the no-transaction band widens, the optimal strategy requires increasingly less rebalancing, so the marginal hedging cost of additional transaction costs diminishes. The DH joint price follows the same pattern, confirming that both approaches capture the same underlying phenomenon.

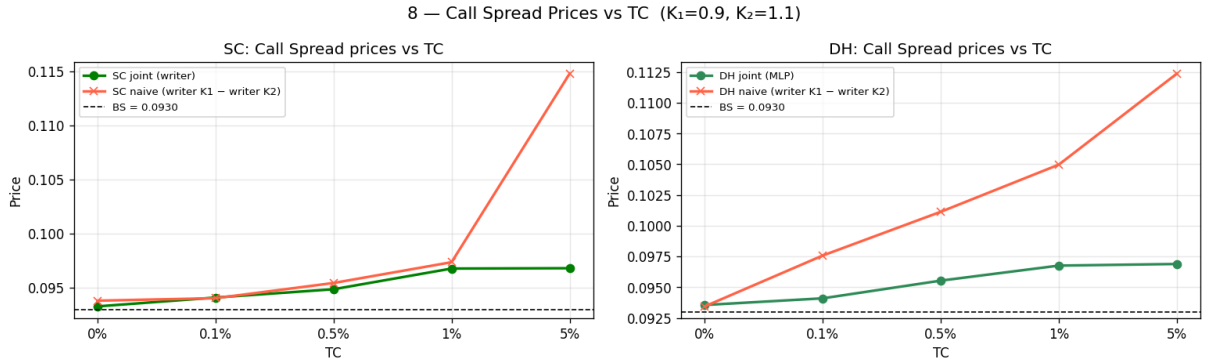


Figure 9: Bull call spread writer prices vs transaction cost level for joint and naive hedging strategies, under SC and DH. The BS reference price is shown as a dashed line.

For the remaining results in this section we focus on the stochastic control solution, which provides the clearest structural illustration of the phenomena of interest. The deep hedging results are qualitatively consistent throughout.

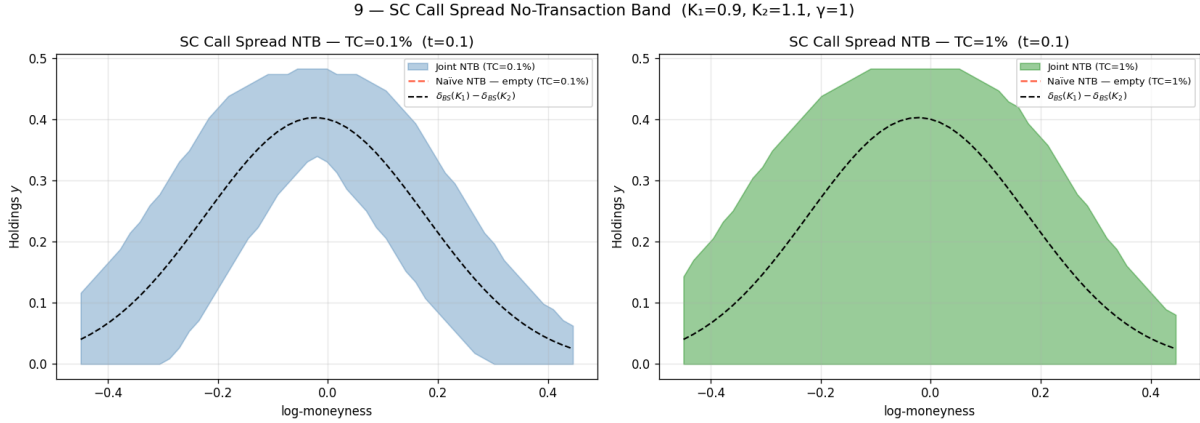
5.9 Bull Call Spread: No-Transaction Band

Figure 10 compares the joint and naive no-transaction bands at $t = 0.1$ for two TC levels. The joint band is centred around the spread delta $\Delta_{BS}(K_1) - \Delta_{BS}(K_2)$ and widens substantially as λ increases from 0.1% to 1%, consistent with the Whalley-Wilmott formula (40) and the gamma cancellation argument of Section 3.

The naive band is the intersection of the writer band for the K_1 leg and the buyer band for the K_2 leg. This is the economically correct decomposition: a trader hedging the two legs independently would target $+\Delta_{BS}(K_1)$ shares for the short K_1 call and $-\Delta_{BS}(K_2)$ shares for the long K_2 call, operating separate no-transaction bands around each target. The intersection of these two bands in terms of the net holding y is the set of positions that simultaneously satisfies both individual band constraints.

The figure reveals that this intersection is empty at both TC levels. This is the direct consequence of the two individual bands being centred at $+\Delta_{BS}(K_1)$ and $-\Delta_{BS}(K_2)$ respectively, which are on opposite sides of the spread delta near the money. A naive hedger

therefore has no region of inaction in the most active part of the state space and is forced into near-continuous rebalancing precisely where the spread's gamma exposure is largest. The joint hedger, by contrast, maintains a wide band centred around $\Delta_{BS}(K_1) - \Delta_{BS}(K_2)$ throughout, trading only when the net holding drifts sufficiently far from the spread delta. This contrast graphically explains both the higher trading frequency and the higher indifference price of the naive approach.



5.10 Bull Call Spread: P&L and Trade Metrics

Figure 11 shows the terminal P&L scatter at $\lambda = 0.5\%$ for SC joint and naive hedging. We focus on this TC level because it lies in the regime where transaction costs are large enough to create a meaningful difference between the two strategies, but below the level at which the joint policy switches to near-inactivity, which would make the comparison less informative. The joint strategy produces a tighter concentration of outcomes around the mean, while the naive strategy exhibits wider dispersion and larger outlier losses, consistent with the excess rebalancing that the empty naive band forces near the money.

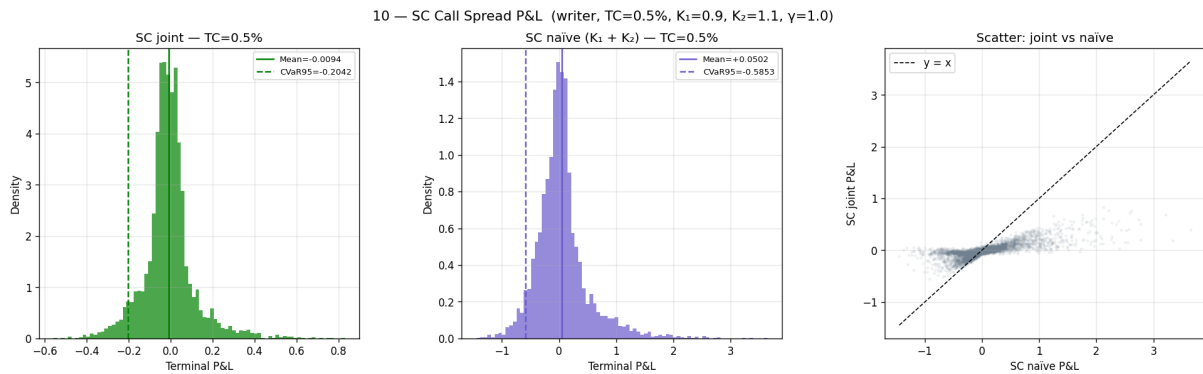


Figure 12 confirms the mechanism quantitatively: the joint strategy trades substantially less frequently than the naive strategy at all TC levels, and the gap widens with λ . At low TC levels the naive strategy trades nearly continuously, directly reflecting the empty naive no-transaction band documented in Section 5.9. This reduced trading activity under joint hedging is the direct cause of the lower indifference prices documented in Section 5.8.

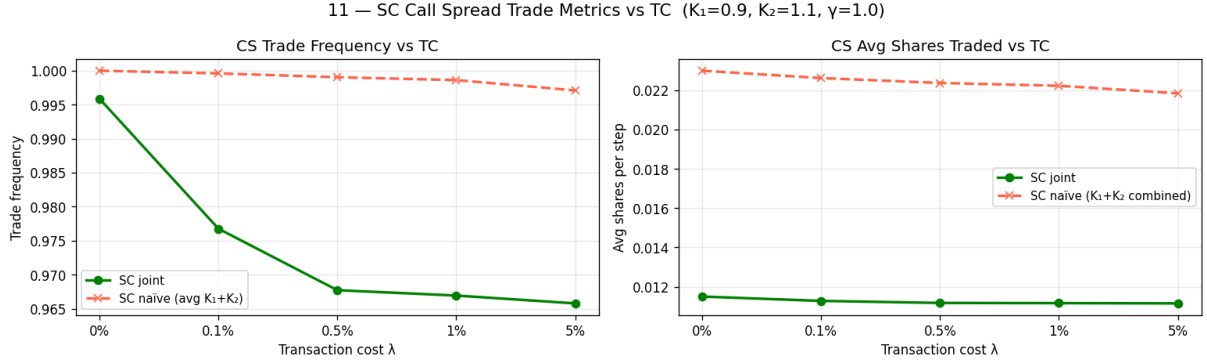


Figure 12: Trade frequency (left) and average shares traded (right) for joint and naive hedging of the bull call spread under SC, vs transaction cost level.

5.11 Summary

Table 5: Qualitative summary of model performance at $\lambda = 1\%$.

Model	Writer Price	Bid-Ask Spread	CVaR (95%)	Band Quality
SC	0.08932	0.01895	Worst	Reference
MLP	0.08639	—	Moderate	None
NTBN- Δ	0.08597	—	Good	Moderate
WW-NTBN	0.08591	0.01339	Best	Best

Table 5 provides a summary of model performance at $\lambda = 1\%$. WW-NTBN achieves the best CVaR among DH models, the tightest bid-ask spread, and the closest match to the SC no-transaction band. SC provides the most conservative writer price but the worst tail risk, reflecting its worst-case optimality guarantees. The MLP, despite having no structural prior, produces competitive prices but trades more and exhibits worse tail behaviour than the band-based architectures. These results support the central conclusion of this paper: embedding the structural knowledge from stochastic control theory into the network architecture, as done in WW-NTBN, improves both the quality of the learned hedging strategy and its quantitative agreement with the analytically optimal solution.

6 Conclusion

This paper has studied the problem of hedging and pricing European options under proportional transaction costs from two complementary perspectives, and shown that the two approaches are more deeply connected than is typically appreciated in the literature.

On the stochastic control side, we derived the HJB quasi-variational inequality governing the optimal strategy under CARA utility, characterised the no-transaction band analytically, and presented the Whalley-Wilmott asymptotic formula as a tractable approximation whose accuracy we verified numerically. The CARA dimension reduction reduces a four-dimensional PDE to a three-dimensional problem, making the dynamic programming scheme computationally viable and enabling exact indifference pricing via a single pair of backward recursions. The bull call spread analysis showed that this framework immediately reveals a phenomenon that is easy to miss in frictionless models: once transaction costs are present, the price of a portfolio of derivatives is strictly less than the sum of the individual prices when hedged jointly, and the gap grows substantially with the transaction cost level.

On the deep hedging side, the main contribution is the WW-NTBN architecture, which uses the Whalley-Wilmott half-width as a structural prior on the no-transaction bandwidth and replaces the hard clamp with a differentiable soft clamp. By embedding the analytical result directly into the network, WW-NTBN begins training with the correct scaling in λ and γ , converges faster than both the MLP and NTBN- Δ baselines, and produces no-transaction bands that quantitatively match the stochastic control solution across all transaction cost levels tested. The intermediate architecture NTBN- Δ , which centres the band around the Black-Scholes delta without imposing a prior on its width, provides a useful model-agnostic baseline when the log-normal assumption is not appropriate.

The numerical comparison reveals a growing divergence between SC and DH prices at high transaction costs that deserves comment. The gap reflects several compounding factors: the SC numerical scheme is restricted to trades of exactly $\pm\Delta y$ shares per step and uses a binomial tree discretisation, both of which introduce approximation errors that grow with λ ; the DH networks are trained on finite batches of simulated paths, which means rare extreme scenarios may not be fully reflected in the learned policy; and the two approaches use fundamentally different time discretisations, making direct quantitative comparison imprecise. Neither approach is uniformly superior. SC provides rigorous optimality guarantees and exact indifference prices under its model assumptions, while deep hedging is more flexible, scales to complex payoffs and portfolios, and does not require solving a high-dimensional PDE. WW-NTBN occupies a deliberate middle ground: it sacrifices some of the model-agnosticism of pure deep hedging in exchange for the structural accuracy of stochastic control, a trade-off that is most valuable when the transaction cost level is moderate and the log-normal assumption is reasonable.

Several directions suggest themselves for future work. The most natural extension is

to apply WW-NTBN beyond the GBM setting, replacing the Black-Scholes gamma with a model-specific hedging sensitivity under stochastic volatility or jump-diffusion dynamics, where the qualitative structure of the no-transaction band is preserved but the frictionless optimum changes. A second direction is to extend the joint hedging framework to larger portfolios of derivatives, where the subadditivity of indifference pricing has potentially significant practical implications for how dealers price and risk-manage structured products. Finally, the WW prior could serve as a warm start for reinforcement learning approaches to hedging, providing a structured initialisation that reduces the exploration cost in the early stages of training.

Appendices

Appendix A. CARA Indirect Utility Derivation

A.1 Without Option Position

We verify the explicit form of the value function under CARA utility $u(W) = 1 - e^{-\gamma W}$ in the frictionless case without an option. We make the ansatz

$$U(t, W) = 1 - e^{-\gamma\varphi(t)W + \psi(t)}, \quad (66)$$

and substitute into the HJB equation (11). The required partial derivatives are

$$U_t = (\gamma\varphi'(t)W - \psi'(t))e^{-\gamma\varphi(t)W + \psi(t)}, \quad (67)$$

$$U_W = \gamma\varphi(t)e^{-\gamma\varphi(t)W + \psi(t)}, \quad (68)$$

$$U_{WW} = -\gamma^2\varphi(t)^2e^{-\gamma\varphi(t)W + \psi(t)}. \quad (69)$$

Substituting the optimal Merton fraction (12) into the HJB equation and dividing through by $e^{-\gamma\varphi(t)W + \psi(t)} \neq 0$, we obtain

$$\gamma\varphi'(t)W - \psi'(t) + \gamma r\varphi(t)W + \frac{(\mu - r)^2}{2\sigma^2} = 0. \quad (70)$$

Matching coefficients in W and in the constant term separately gives the system

$$\varphi'(t) + r\varphi(t) = 0, \quad (71)$$

$$\psi'(t) = \frac{(\mu - r)^2}{2\sigma^2}. \quad (72)$$

With boundary conditions $\varphi(T) = 1$ and $\psi(T) = 0$, the solutions are

$$\varphi(t) = e^{r(T-t)}, \quad \psi(t) = \frac{(\mu - r)^2}{2\sigma^2}(t - T), \quad (73)$$

giving the value function (15). The optimal fraction (16) follows immediately from (12).

A.2 Dimension Reduction under Transaction Costs

We prove Proposition 2.3. The argument proceeds in three steps: we first express X_T explicitly as a function of X_t and the controls, then use the CARA structure to factor out the X_t dependence from the value function, and finally derive the reduced quasi-variational inequality for Q .

Step 1: Solution of the cash dynamics. The cash balance satisfies $dX_t = rX_t dt - (1 + \lambda)S_t l_t dt + (1 - \lambda)S_t m_t dt$. Solving this linear SDE gives

$$X_T = \frac{X_t}{\delta(t, T)} + \int_t^T \frac{1}{\delta(s, T)} [(1 - \lambda)S_s dm_s - (1 + \lambda)S_s dl_s], \quad (74)$$

where $\delta(t, T) = e^{-r(T-t)}$. The first term is the initial cash balance compounded at the risk-free rate. The integral collects the discounted net proceeds from all trading between t and T . Crucially, only the first term depends on X_t ; the integral depends on the controls (l, m) and the stock price path, but not on the initial cash balance.

Step 2: Factorisation of the terminal utility. Substituting (74) into the terminal utility $u(X_T + y_T S_T - \lambda S_T |y_T| - \varphi(S_T))$ and writing $\mathcal{G}^{l, m}$ for all terms in the argument that do not depend on X_t ,

$$\begin{aligned} & u(X_T + y_T S_T - \lambda S_T |y_T| - \varphi(S_T)) \\ &= 1 - \exp\left(-\gamma \left(\frac{X_t}{\delta(t, T)} + \mathcal{G}^{l, m}\right)\right) \\ &= 1 - e^{-\gamma X_t / \delta(t, T)} \cdot e^{-\gamma \mathcal{G}^{l, m}}, \end{aligned} \quad (75)$$

where the last step uses $e^{-(a+b)} = e^{-a} \cdot e^{-b}$. This is the key step: the CARA utility converts the additive separation of X_t in the terminal wealth into a multiplicative separation in the utility. The factor $e^{-\gamma X_t / \delta(t, T)}$ is $\mathcal{F}_t$ -measurable and strictly positive, so it is a constant with respect to both the conditional expectation and the supremum over future controls (l, m) .

Taking the supremum over (l, m) and the conditional expectation,

$$\begin{aligned} U(t, S, y, X) &= \sup_{(l, m)} \mathbb{E}_{t, S, y, X} [1 - e^{-\gamma X_t / \delta(t, T)} \cdot e^{-\gamma \mathcal{G}^{l, m}}] \\ &= 1 - e^{-\gamma X / \delta(t, T)} \inf_{(l, m)} \mathbb{E}_{t, S, y} [e^{-\gamma \mathcal{G}^{l, m}}], \end{aligned} \quad (76)$$

where the sup over (l, m) becomes an inf because maximising $1 - e^{-\gamma X / \delta} \cdot Z$ over (l, m) is equivalent to minimising $Z = e^{-\gamma \mathcal{G}^{l, m}}$ over (l, m) , since the prefactor $e^{-\gamma X / \delta} > 0$ is independent of the controls. Setting

$$Q(t, y, S) = \inf_{(l, m) \geq 0} \mathbb{E}_{t, y, S} \left[\exp\left(-\gamma \int_t^T \frac{(1 - \lambda)S_s dm_s - (1 + \lambda)S_s dl_s}{\delta(s, T)}\right) e^{-\gamma(y_T S_T - \lambda S_T |y_T| - \varphi(S_T))} \right], \quad (77)$$

equation (76) gives the factorisation $U(t, S, y, X) = 1 - e^{-\gamma X / \delta(t, T)} Q(t, y, S)$. Since the infimum is taken over (t, y, S) alone with no reference to X , the function Q does not depend on X .

Step 3: Derivation of the reduced quasi-variational inequality. We now substitute the factorised form $U = 1 - e^{-\gamma X/\delta} Q$ into the quasi-variational inequality $\max\{A, B, C\} = 0$ and derive the equation satisfied by Q . The required partial derivatives are

$$U_X = \frac{\gamma}{\delta(t, T)} e^{-\gamma X/\delta} Q, \quad (78)$$

$$U_y = -e^{-\gamma X/\delta} Q_y, \quad (79)$$

$$U_S = -e^{-\gamma X/\delta} Q_S, \quad (80)$$

$$U_{SS} = -e^{-\gamma X/\delta} Q_{SS}, \quad (81)$$

$$U_t = -e^{-\gamma X/\delta} \left(Q_t - \frac{\gamma r X}{\delta(t, T)} Q \right). \quad (82)$$

Substituting into $A = U_y - (1 + \lambda) S U_X$:

$$A = -e^{-\gamma X/\delta} \left(Q_y + \frac{\gamma(1 + \lambda) S}{\delta(t, T)} Q \right). \quad (83)$$

Substituting into $B = -U_y + (1 - \lambda) S U_X$:

$$B = -e^{-\gamma X/\delta} \left(-Q_y - \frac{\gamma(1 - \lambda) S}{\delta(t, T)} Q \right). \quad (84)$$

Substituting into $C = -U_t - \mu S U_S - \frac{1}{2} \sigma^2 S^2 U_{SS} - r X U_X$, and noting that the $r X U_X$ term cancels the $\gamma r X / \delta \cdot Q$ term arising from U_t :

$$C = -e^{-\gamma X/\delta} \left(-Q_t - \mu S Q_S - \frac{1}{2} \sigma^2 S^2 Q_{SS} \right). \quad (85)$$

In each case A, B, C equal $-e^{-\gamma X/\delta}$ times a function of (t, y, S) alone. Since $-e^{-\gamma X/\delta} < 0$, the condition $\max\{A, B, C\} = 0$ is equivalent to

$$\min \left\{ Q_y + \frac{\gamma(1 + \lambda) S}{\delta(t, T)} Q, -Q_y - \frac{\gamma(1 - \lambda) S}{\delta(t, T)} Q, -Q_t - \mu S Q_S - \frac{1}{2} \sigma^2 S^2 Q_{SS} \right\} = 0, \quad (86)$$

which is the reduced quasi-variational inequality (36). The terminal condition $Q(T, y, S) = \exp(-\gamma(yS - \lambda S|y| - \varphi(S)))$ follows directly from (33) and the factorisation. This completes the proof of Proposition 2.3.

Appendix B. Sketch of the Whalley–Wilmott Asymptotic Derivation

We outline the logic of the matched asymptotic expansion of Whalley and Wilmott (1997), which reduces the three-dimensional free boundary problem for Q to the closed-form bandwidth formula (40). The emphasis is on the structure of the argument rather than the algebraic details, which can be found in Section 3 of the original paper.

Setting

The reduced variational inequality (36) partitions the (S, y) plane at each time t into three regions: buy, sell, and no-transaction. In the buy and sell regions, the active conditions $A = 0$ and $B = 0$ are first-order in y , so for each fixed (S, t) they can be integrated as ODEs in y . This gives the *exact* general solutions

$$Q^{\text{buy}} = \exp\left(-\frac{\gamma(1+\lambda)Sy}{\delta} + H^-(S, t; \lambda)\right), \quad (87)$$

$$Q^{\text{sell}} = \exp\left(-\frac{\gamma(1-\lambda)Sy}{\delta} + H^+(S, t; \lambda)\right), \quad (88)$$

where $\delta = e^{-r(T-t)}$ and $H^\pm$ are undetermined functions of (S, t) . Both solutions are *exponential-linear* in y : the y -dependence of $\log Q$ is exactly $-\gamma(1 \pm \lambda)Sy/\delta$.

In the no-transaction region, Q satisfies the full PDE $Q_t + \mu SQ_S + \frac{1}{2}\sigma^2 S^2 Q_{SS} = 0$, and must be joined to (87)–(88) at the unknown free boundaries with Q , Q_y , and Q_{yy} all continuous.

The expansion in the no-transaction region

It is natural to work with $\log Q$ rather than Q itself. Since the no-transaction region is a narrow band around the frictionless optimum $y^*(S, t)$, we introduce a rescaled variable Y defined by $y = y^* + \lambda^{1/3}Y$, so that $Y = O(1)$ inside the band. The leading y -dependent term $-\gamma Sy^*/\delta$ in the expansion below is forced by continuity: both (87) and (88) contain $-\gamma Sy/\delta$ at $\lambda = 0$, and since $y = y^* + \lambda^{1/3}Y$ this splits into $-\gamma Sy^*/\delta$ at $O(1)$ and $-\lambda^{1/3}\gamma SY/\delta$ at $O(\lambda^{1/3})$. We then write

$$\log Q = -\frac{\gamma S y^*}{\delta} + H_0(S, t) - \lambda^{1/3} \frac{\gamma SY}{\delta} + \lambda^{1/3} H_1(S, t) + \lambda^{2/3} H_2(S, t) + \lambda H_3(S, t) + \dots$$

where terms depending on both Y and (S, t) also appear at higher orders (starting at $O(\lambda^{4/3})$). The exponent $1/3$ is not chosen arbitrarily: it is the unique value for which the interior PDE and the boundary matching contribute at the same asymptotic order, as can be verified by tracking powers of λ through the expansion (see [Whalley and Wilmott \(1997\)](#), Section 3, for the detailed calculation). An economic interpretation of this scaling is given at the end of this appendix.

The functions $H_0, H_1, H_2, \dots$ and the higher-order Y -dependent terms are determined by substituting the expansion into the no-transaction PDE and collecting powers of $\lambda^{1/3}$. A key subtlety is that the PDE is written in derivatives at fixed y , not fixed Y . Since Y depends on S and t through y^* , the chain rule introduces factors of y_S^* (the option's gamma) whenever S -derivatives act on Y -dependent terms, coupling different orders of the expansion.

What each order determines

$O(1)$. Determines $H_0(S, t)$ via a PDE relating it to y^* .

$O(\lambda^{1/3})$. This order contains terms proportional to Y and terms independent of Y ; each group must vanish separately. The Y -dependent part yields the frictionless optimal holding: $y^*(S, t) = \partial V_0 / \partial S + \delta(\mu - r) / (\gamma S \sigma^2)$, where $V_0 = (\delta / \gamma) H_0$ satisfies the Black–Scholes equation. The Y -independent part gives an equation for H_1 , which turns out to be identically zero: the first non-trivial correction is at $O(\lambda^{2/3})$, not $O(\lambda^{1/3})$.

$O(\lambda^{2/3})$: the bandwidth. This is the critical order. Through the chain-rule mechanism described above, S -derivatives acting on higher-order Y -dependent terms bring them down to this order: specifically, the second S -derivative of a Y -dependent function at $O(\lambda^{4/3})$ in $\log Q$ picks up a factor $\lambda^{-2/3}$ from two applications of $y_S^* \cdot \lambda^{-1/3} \partial / \partial Y$, contributing at $O(\lambda^{2/3})$ in the PDE. This yields an ODE in Y with an inhomogeneous term proportional to Y^2 , whose solution is a polynomial in Y with coefficients depending on $H_2(S, t)$ and on $y_S^* = \partial y^* / \partial S$.

The free constants and the band boundaries Y^+ , Y^- are pinned down by the matching conditions:

- **Continuity of Q_y :** the slope $\partial(\log Q) / \partial y$ must match the buy/sell solutions at each boundary.
- **Continuity of Q_{yy} :** since $\log Q$ is linear in y in the buy/sell regions, $(\log Q)_{yy} = 0$ there, forcing the second Y -derivative of the polynomial to vanish at both boundaries.

These conditions together force (i) the band to be *symmetric*: $Y^+ = Y^-$, and (ii) the half-width:

$$Y^+ = \left(\frac{3 S \delta (y_S^*)^2}{2\gamma} \right)^{1/3}. \quad (89)$$

The actual half-width in the original y -variable is $h = \lambda^{1/3} Y^+$.

The final formula

With the option liability present and $\mu = r$, the speculative term in y^* vanishes and $y_S^* = \Gamma_{BS}(t, S)$. Substituting into (89):

$$h_{WW}(t, S) = \left(\frac{3 \lambda \delta(t, T) S \Gamma_{BS}(t, S)^2}{2\gamma} \right)^{1/3},$$

which is the formula (40).

Why $\lambda^{1/3}$: an economic interpretation

The investor balances two costs. The *suboptimality cost* of holding $y \neq y^*$ is quadratic in the deviation h (by concavity of the value function), contributing a running cost of order h^2 per unit time. The *transaction cost* of rebalancing is λ per unit traded, but a wider band is hit less often: the expected cost per unit time is of order λ/h . The total running cost is therefore $Ah^2 + B\lambda/h$, minimised at $h \propto \lambda^{1/3}$, with minimum value scaling as $\lambda^{2/3}$, which is why the leading correction to the option price is at order $\lambda^{2/3}$, not λ .

Remark B.1. *The full derivation, including the smoothness conditions, the explicit polynomial form of the Y -dependent terms, and the inhomogeneous Black-Scholes equation satisfied by the $O(\lambda^{2/3})$ price correction, is given in Section 3 of [Whalley and Wilmott \(1997\)](#).*

Appendix C. Soft Clamp Gradient

We verify that the soft clamp (62) provides non-zero gradients everywhere. Recall that

$$\text{softclamp}(x, \ell, u) = m + h \cdot \tanh\left(\beta \cdot \frac{x - m}{h}\right), \quad (90)$$

where $m = (\ell + u)/2$ and $h = (u - \ell)/2 > 0$. The derivative with respect to x is

$$\frac{\partial}{\partial x} \text{softclamp}(x, \ell, u) = \beta \left(1 - \tanh^2\left(\beta \cdot \frac{x - m}{h}\right)\right) = \beta \operatorname{sech}^2\left(\beta \cdot \frac{x - m}{h}\right). \quad (91)$$

Since $\operatorname{sech}^2(\cdot) > 0$ everywhere and $\beta > 0$, the gradient (91) is strictly positive for all $x \in \mathbb{R}$. In particular, when $x \ll \ell$ or $x \gg u$, the gradient is small but non-zero, of order $4\beta e^{-2\beta|x-m|/h}$, providing a learning signal even when the previous hedge is far outside the band.

By contrast, the hard clamp $\text{clamp}(x, \ell, u)$ has zero gradient for $x < \ell$ and $x > u$, which removes the gradient signal for any trajectory where the position exits the band. As $\beta \rightarrow \infty$, the soft clamp converges pointwise to the hard clamp while maintaining non-zero gradients for all finite β . In our experiments we use $\beta = 10$, which provides a close approximation to the hard clamp within the band while preserving meaningful gradient flow outside it.

References

- Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.
- Phelim P. Boyle and Ton Vorst. Option replication in discrete time with transaction costs. *Journal of Finance*, 47(1):271–293, 1992.
- Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19(8):1271–1291, 2019.
- Mark H. A. Davis, Vassilios G. Panas, and Thaleia Zariphopoulou. European option pricing with transaction costs. *SIAM Journal on Control and Optimization*, 31(2):470–493, 1993.
- Chanaka Edirisinghe, Vasanttilak Naik, and Raman Uppal. Optimal replication of options with transactions costs and trading restrictions. *Journal of Financial and Quantitative Analysis*, 28(1):117–138, 1993.
- Sergei Fedotov and Sergei Mikhailov. Option pricing for incomplete markets via stochastic optimization: transaction costs, adaptive control and forecast. *International Journal of Theoretical and Applied Finance*, 4(01):179–195, 2001.
- Hans Föllmer and Alexander Schied. *Stochastic Finance: An Introduction in Discrete Time*. Walter de Gruyter, Berlin, 2002.
- Vicky Henderson and David Hobson. Utility indifference pricing: an overview. In René Carmona, editor, *Indifference Pricing: Theory and Applications*, pages 44–74. Princeton University Press, 2004.
- Stewart Hodges and Anthony Neuberger. Optimal replication of contingent claims under transaction costs. *Review of Futures Markets*, 8(2):222–239, 1989.
- Kurt Hornik. Approximation capabilities of multilayer feedforward networks. *Neural Networks*, 4(2):251–257, 1991.
- Blanka Horvath, Josef Teichmann, and Zan Zuric. Deep hedging under rough volatility. *Risks*, 9(7):138, 2021.
- Shota Imaki, Kentaro Imajo, Katsuya Ito, Kentaro Minami, and Kei Nakagawa. No-transaction band network: A neural network architecture for efficient deep hedging. *Journal of Financial Data Science*, 5(2):84–99, 2023. First circulated as arXiv:2103.01775, 2021.

- Jan Kallsen and Johannes Muhle-Karbe. Option pricing and hedging with small transaction costs. *Mathematical Finance*, 25(4):702–723, 2015.
- Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *Proceedings of the International Conference on Learning Representations (ICLR)*, 2015.
- Chi Lee. *Option Pricing in the Presence of Transactions Costs*. PhD thesis, Master thesis at New York University, Dept. of Mathematics, 2000.
- Hayne E. Leland. Option pricing and replication with transaction costs. *Journal of Finance*, 40(5):1283–1301, 1985.
- Robert C. Merton. Lifetime portfolio selection under uncertainty: The continuous-time case. *Review of Economics and Statistics*, 51(3):247–257, 1969.
- Wulin Suo. Hedging options with transaction costs. In *Stochastic Processes, Optimization, and Control Theory: Applications in Financial Engineering, Queueing Networks, and Manufacturing Systems: A Volume in Honor of Suresh Sethi*, pages 223–247. Springer, 2006.
- A. Elizabeth Whalley and Paul Wilmott. An asymptotic analysis of an optimal hedging model for option pricing with transaction costs. *Mathematical Finance*, 7(3):307–324, 1997.